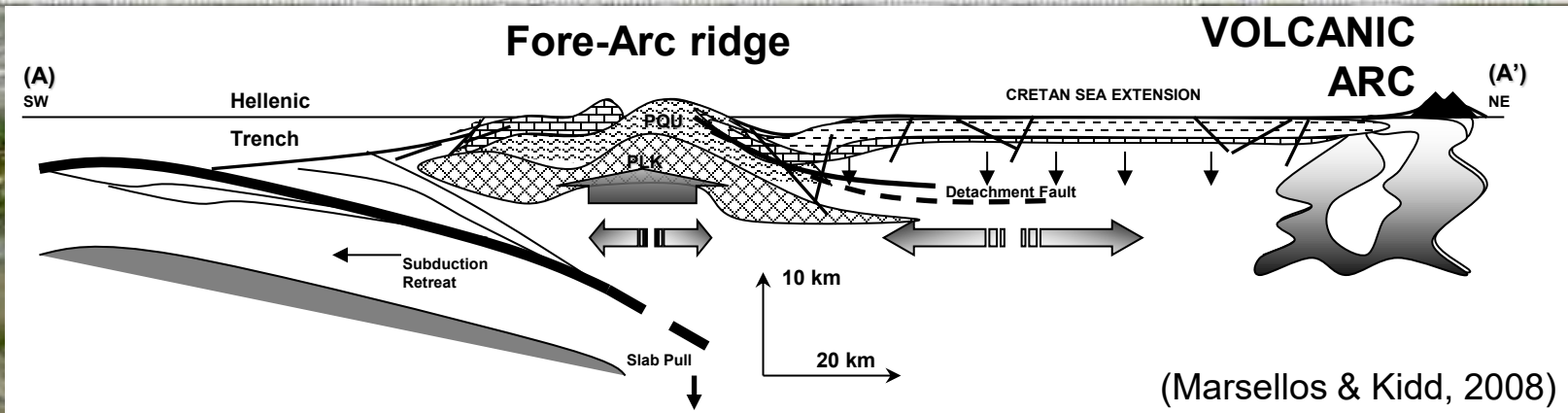


***An application of GIS on structural data
from metamorphic rocks of Santorini Island***



***Dr. Antonios Marsellos
University of Brighton***

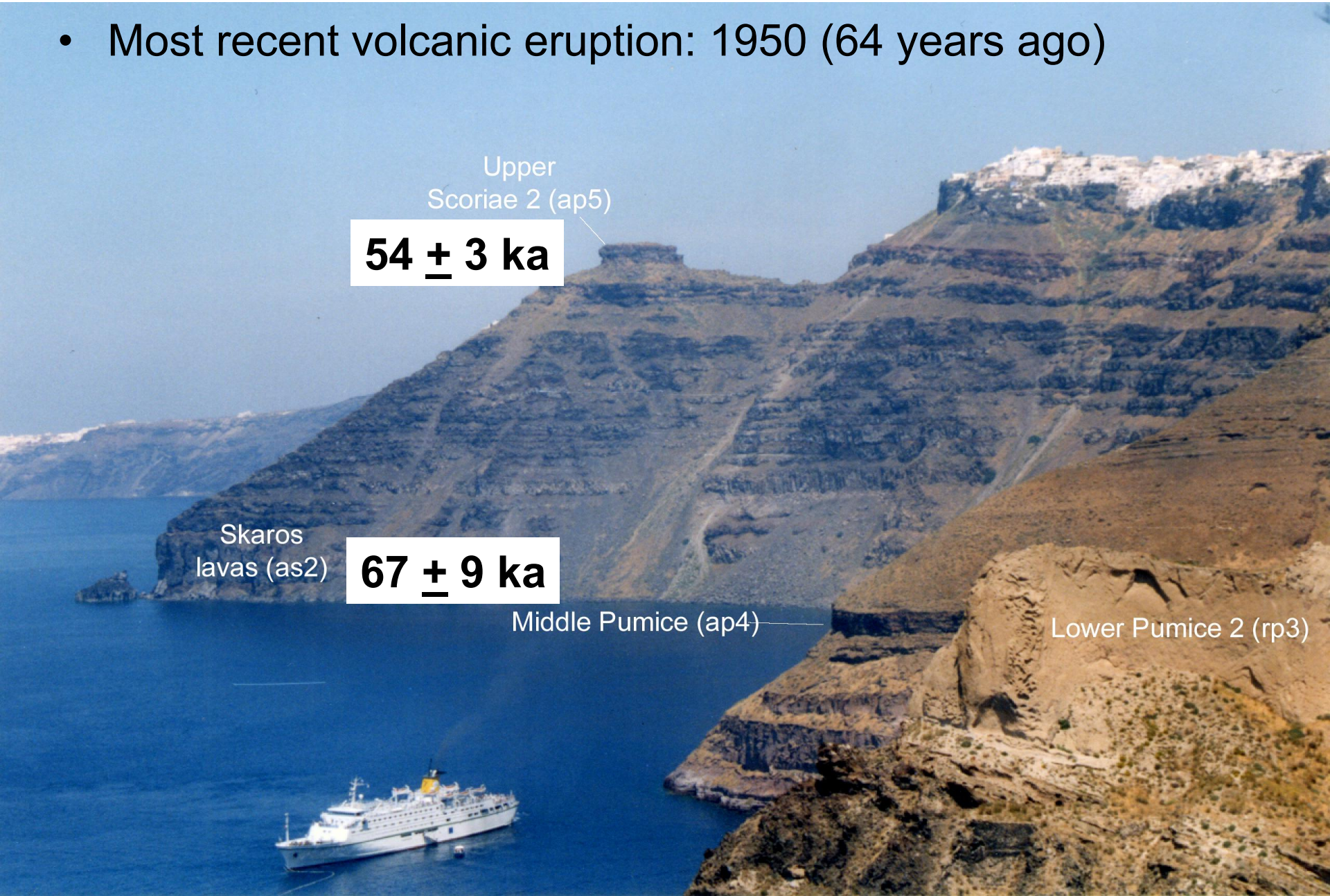


Volcanic Arc in South Aegean



Oldest Santorini Volcanic rock: 650 ka

- Most recent volcanic eruption: 1950 (64 years ago)



Upper
Scoriae 2 (ap5)

54 ± 3 ka

Skaros
lavas (as2)

67 ± 9 ka

Middle Pumice (ap4)

Lower Pumice 2 (rp3)

Questions to answer

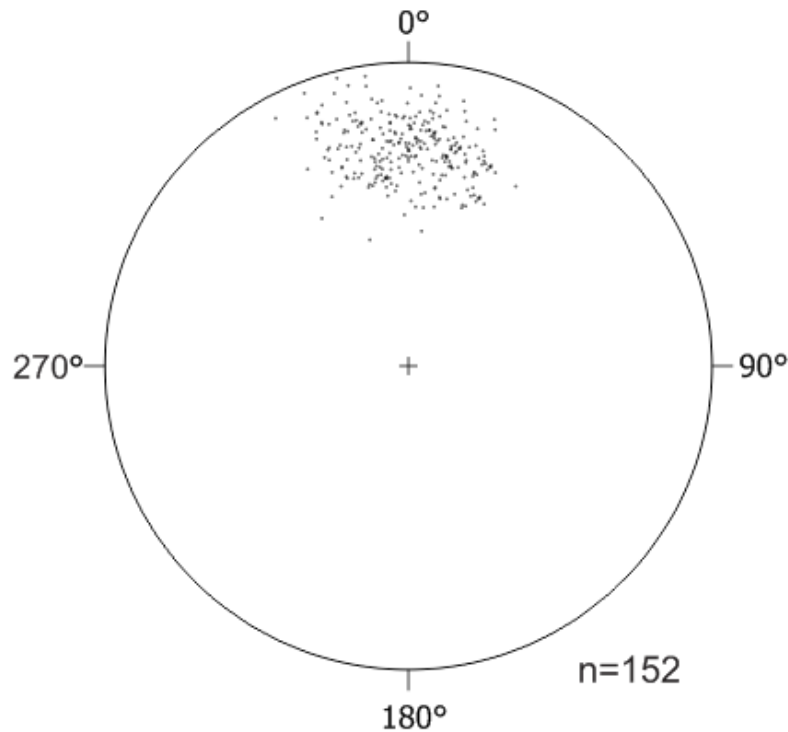
- Has the metamorphic rock unit(s) been reset?
- Does it show arc-parallel or arc-normal extension?
- How could we improve our field time, mapping and sampling



Miocene Extension (15-9 Ma)

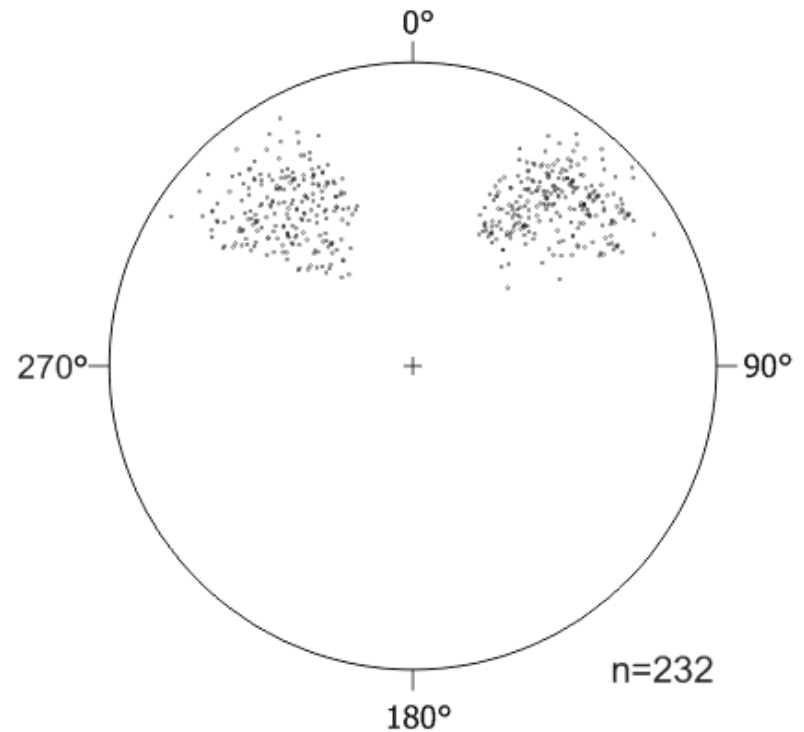
- Lineation measurements

Volcanic Arc & Back-Arc



Equal angle projection, lower hemisphere

Forearc



Equal angle projection, lower hemisphere

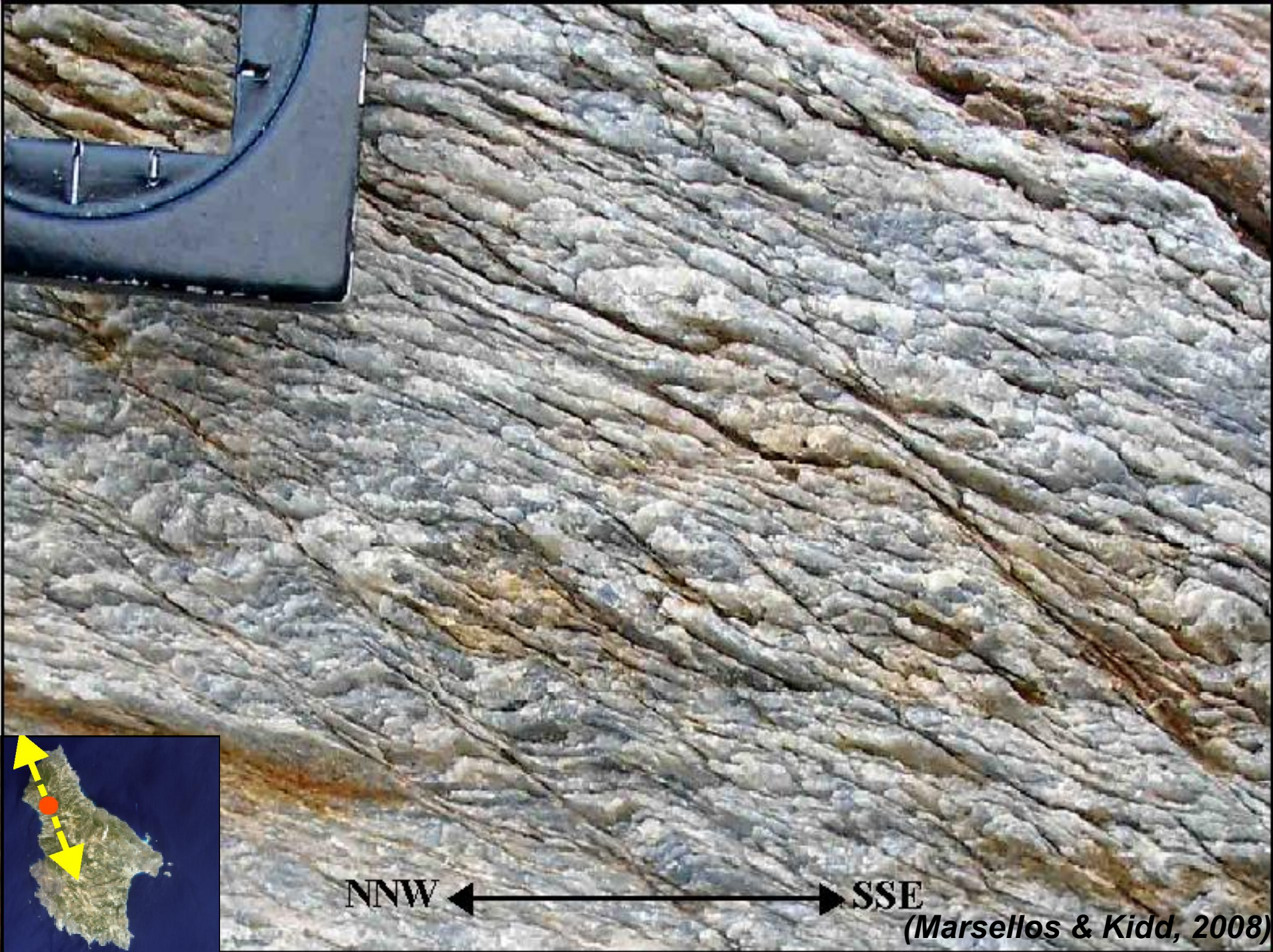


Volcanic Arc

Forearc

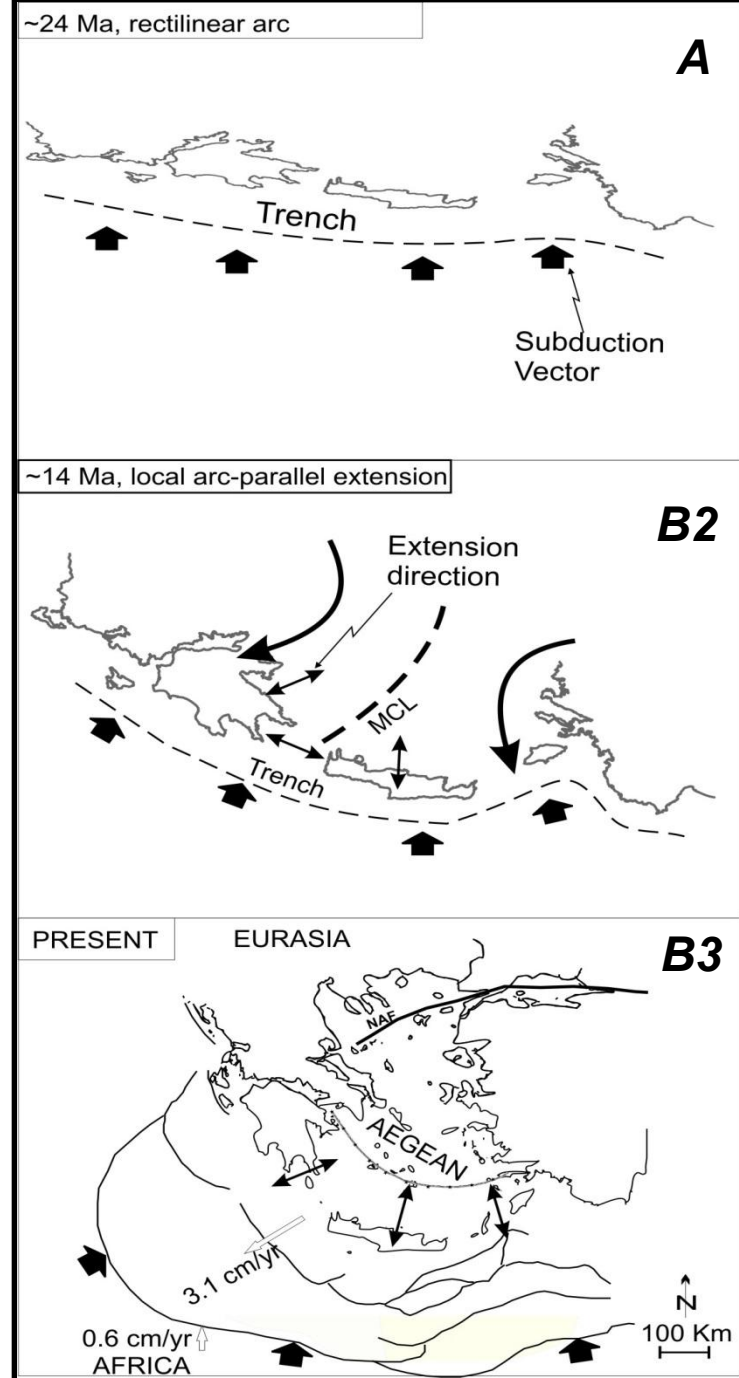
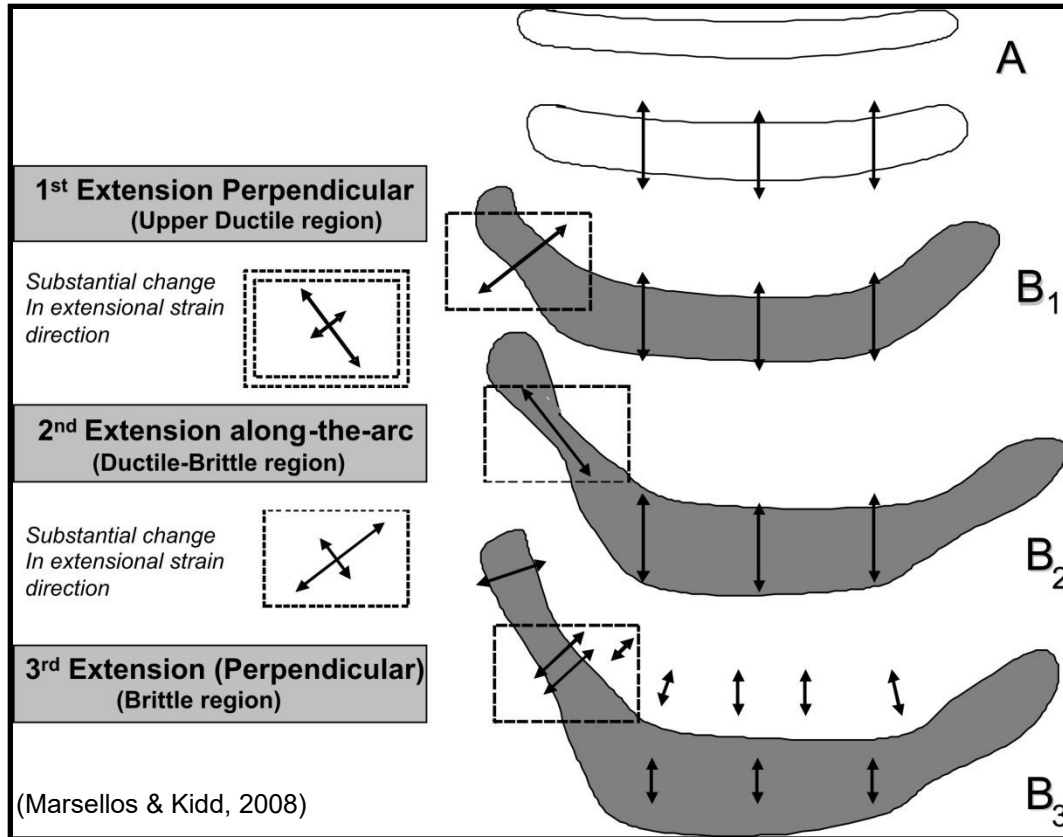


EXTENSION ALONG-THE-ARC (NW-SE)



Forearc extension

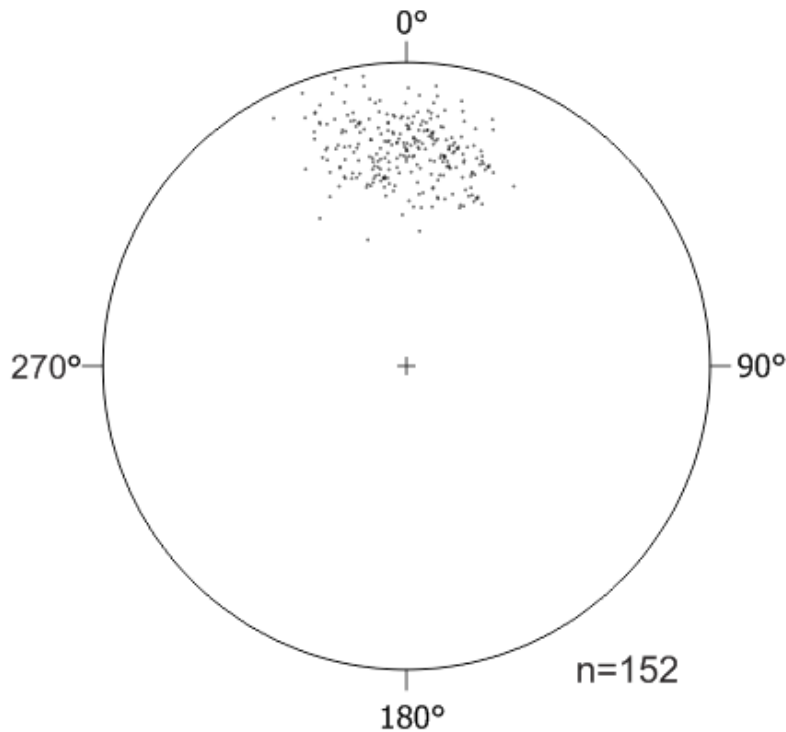
- Arc-normal & Arc-parallel



How many lineation populations?

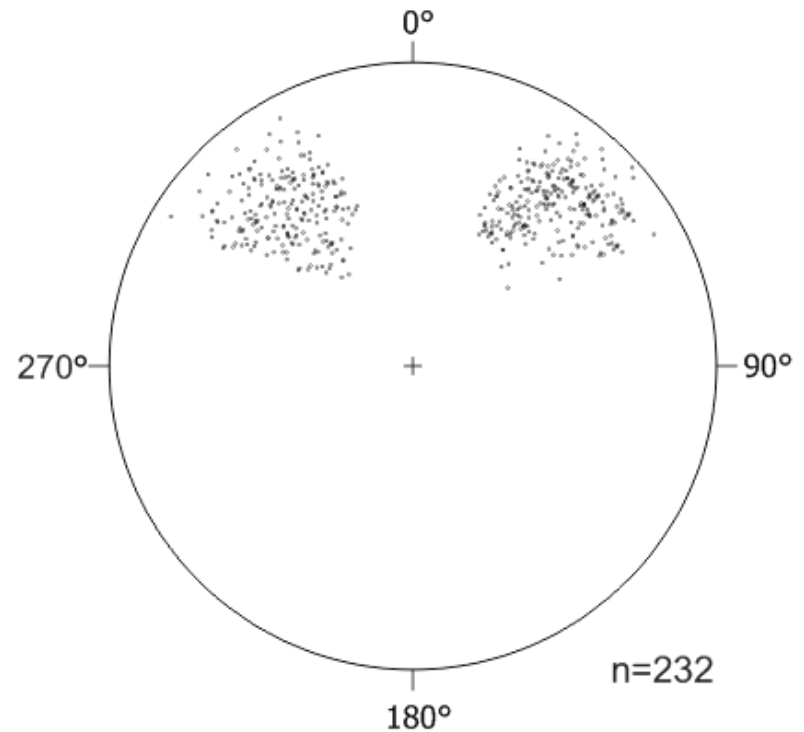
- Lineation measurements

Volcanic Arc & Back-Arc



Equal angle projection, lower hemisphere

Forearc

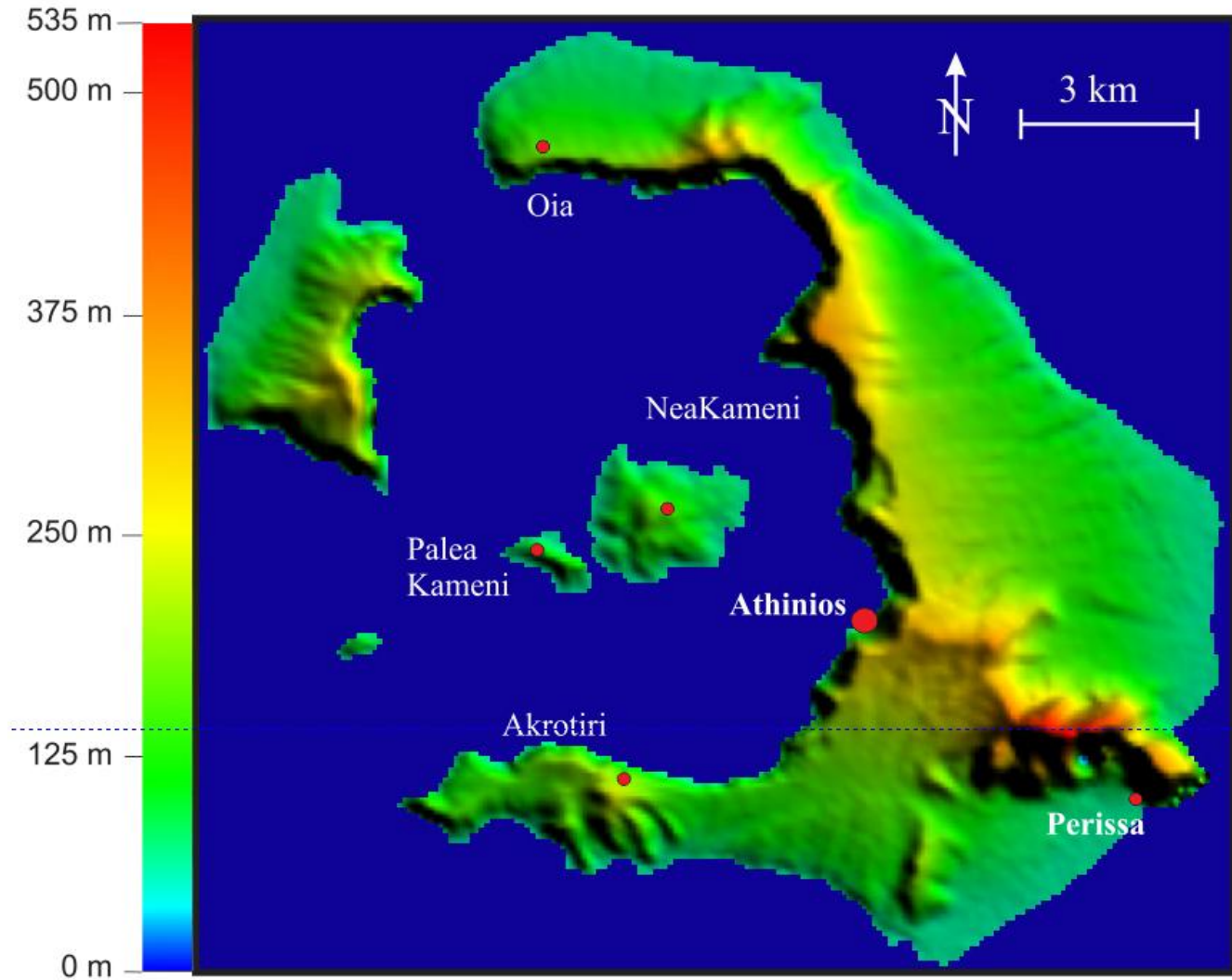


Equal angle projection, lower hemisphere

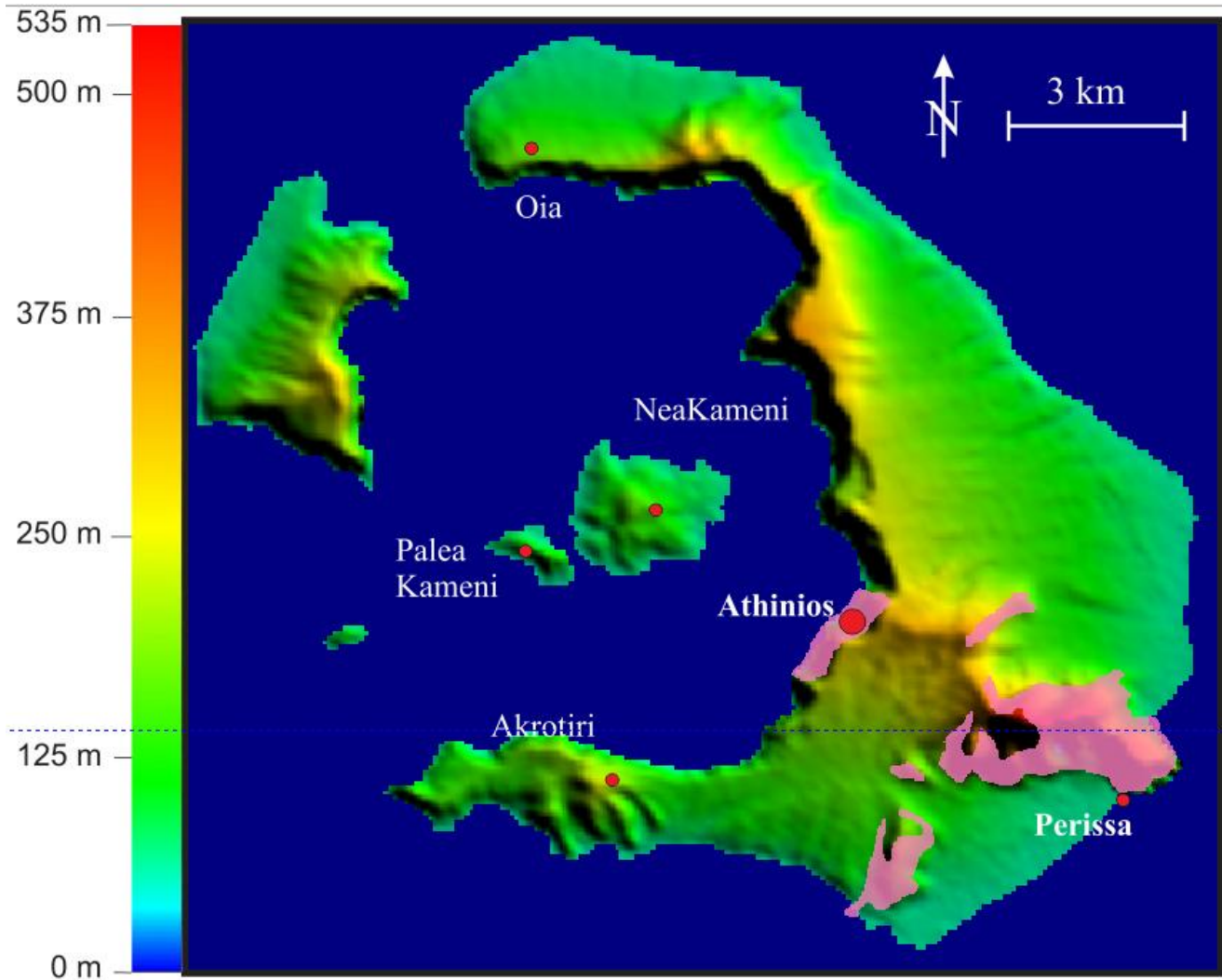
Looking for a suitable study area



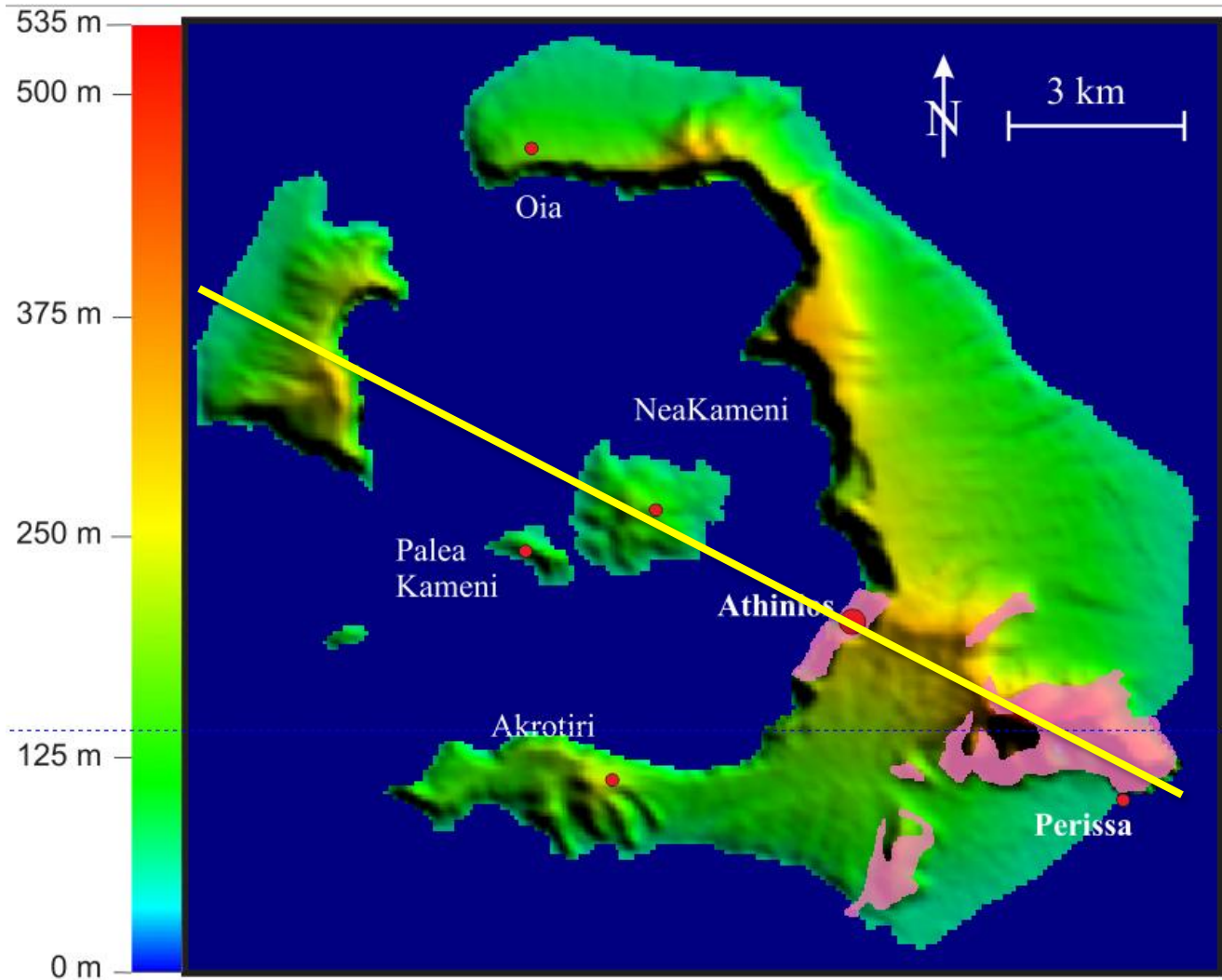
Digital Elevation Model (DEM)



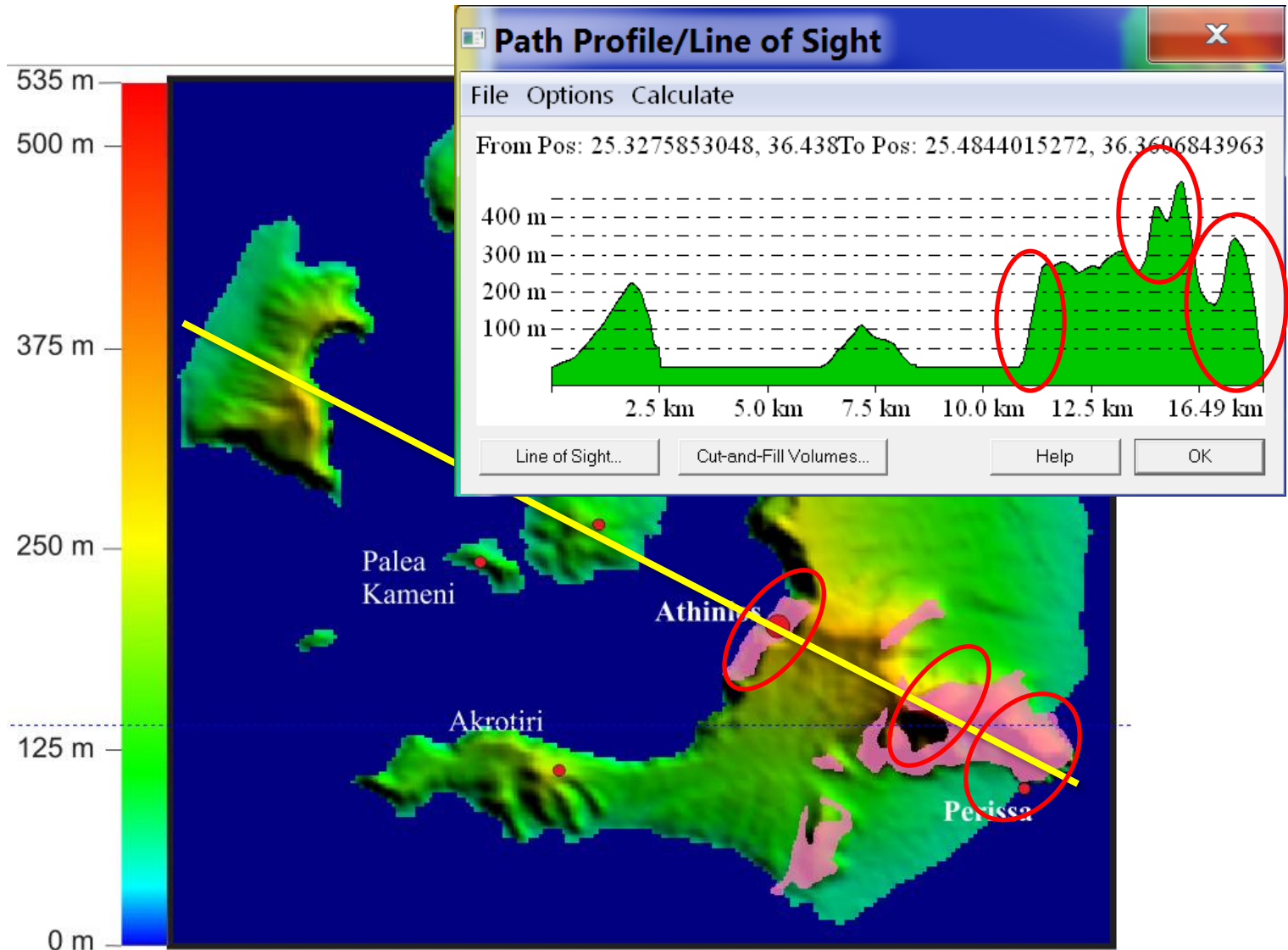
Metamorphic rocks



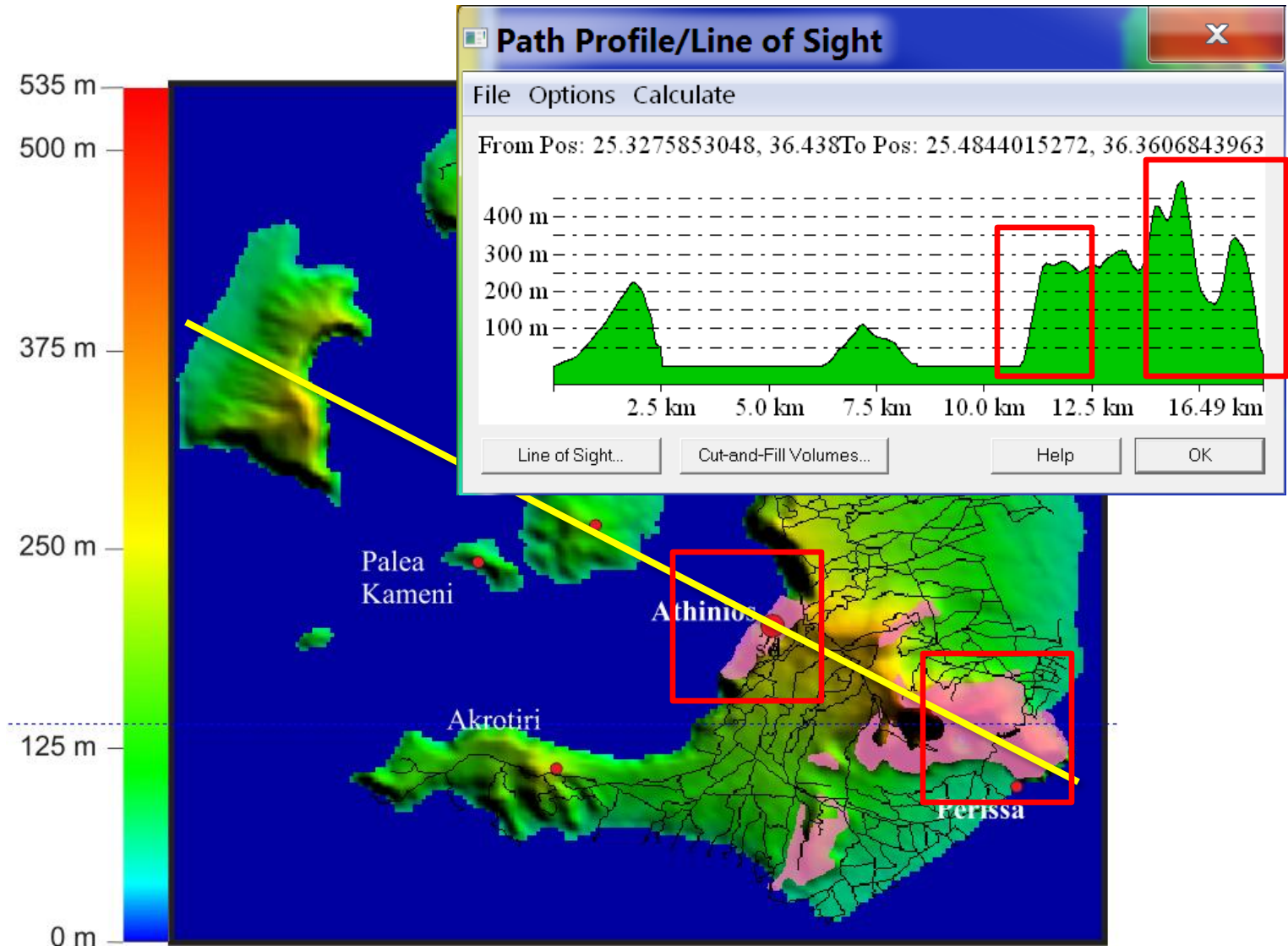
Quick Topographic profile



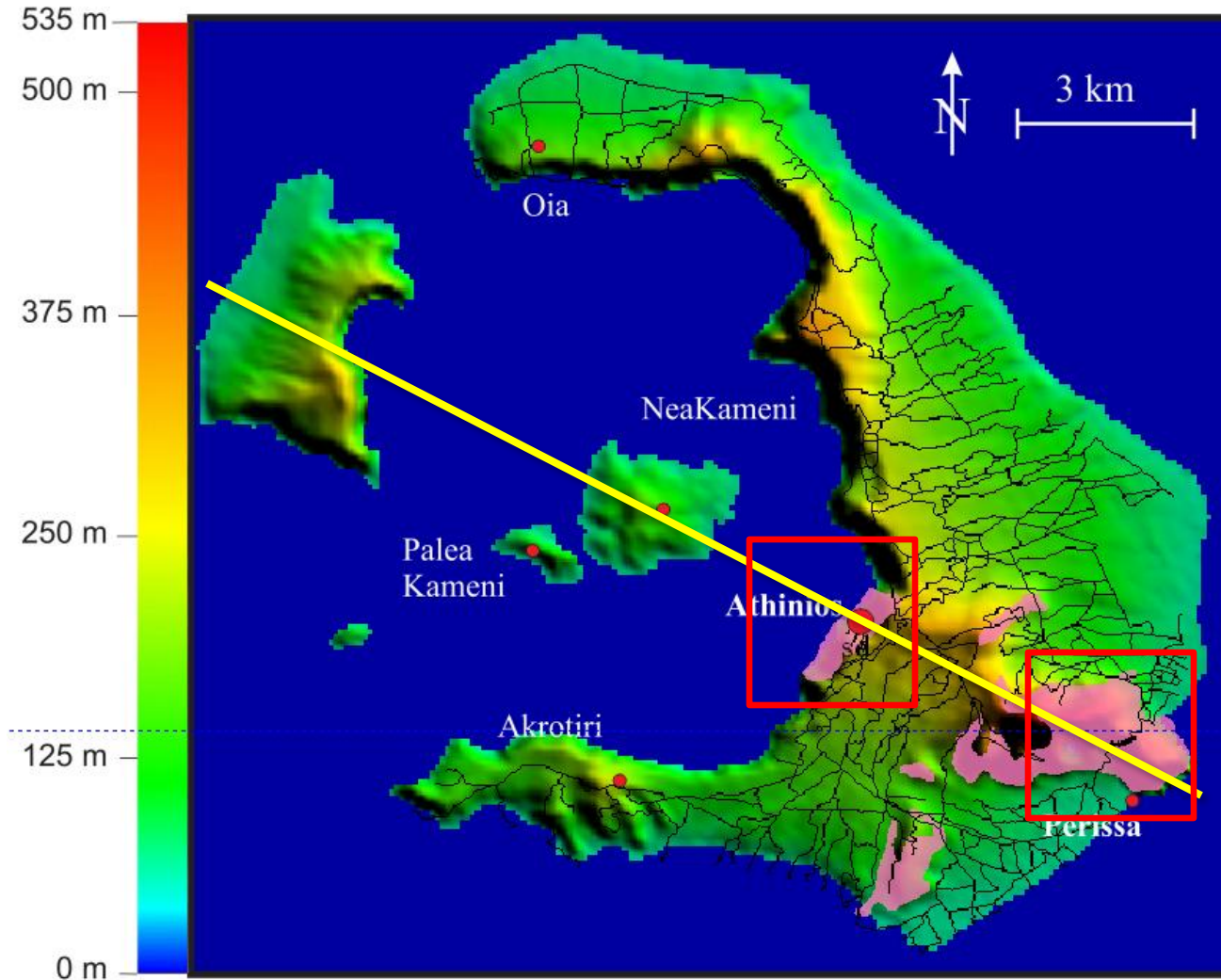
Quick Topographic profile



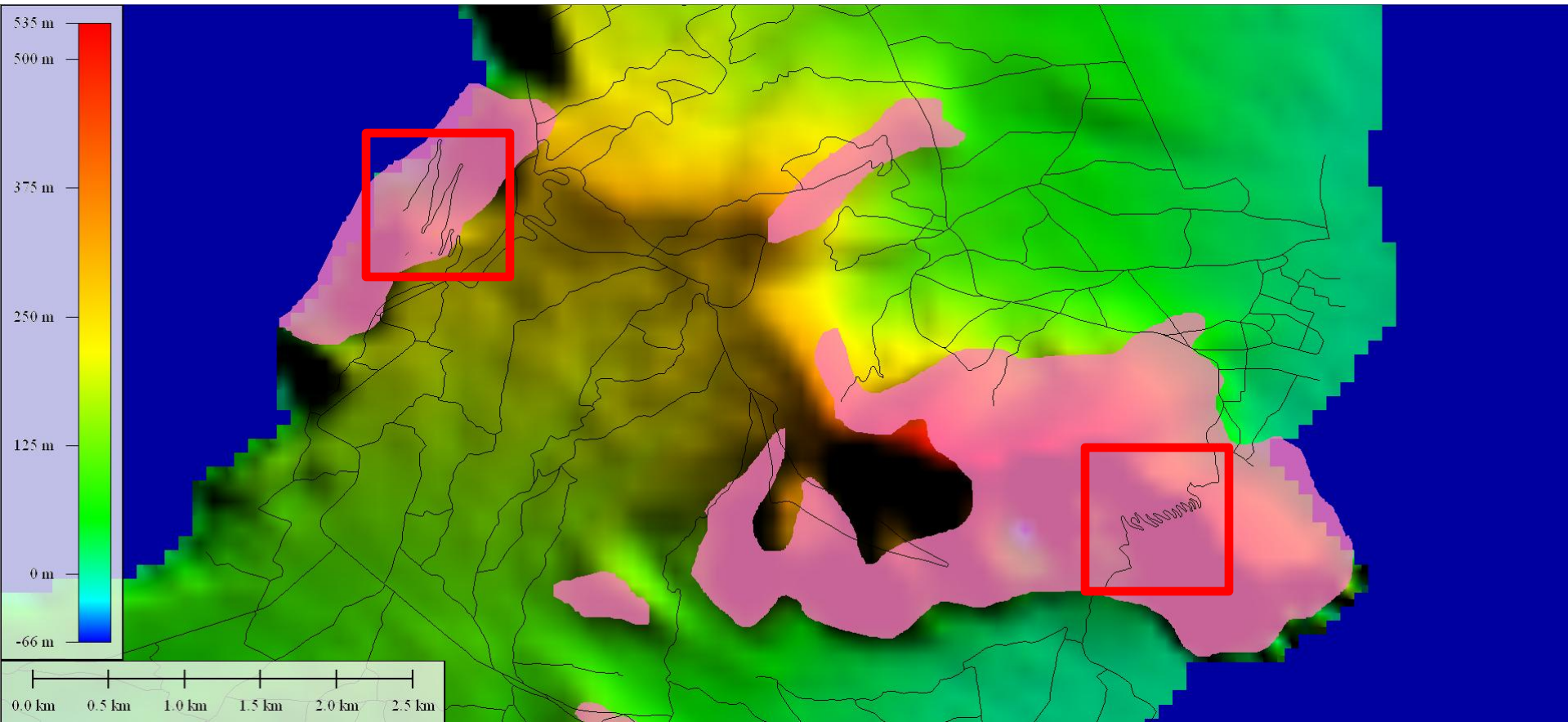
Access to outcrops (add Roads)



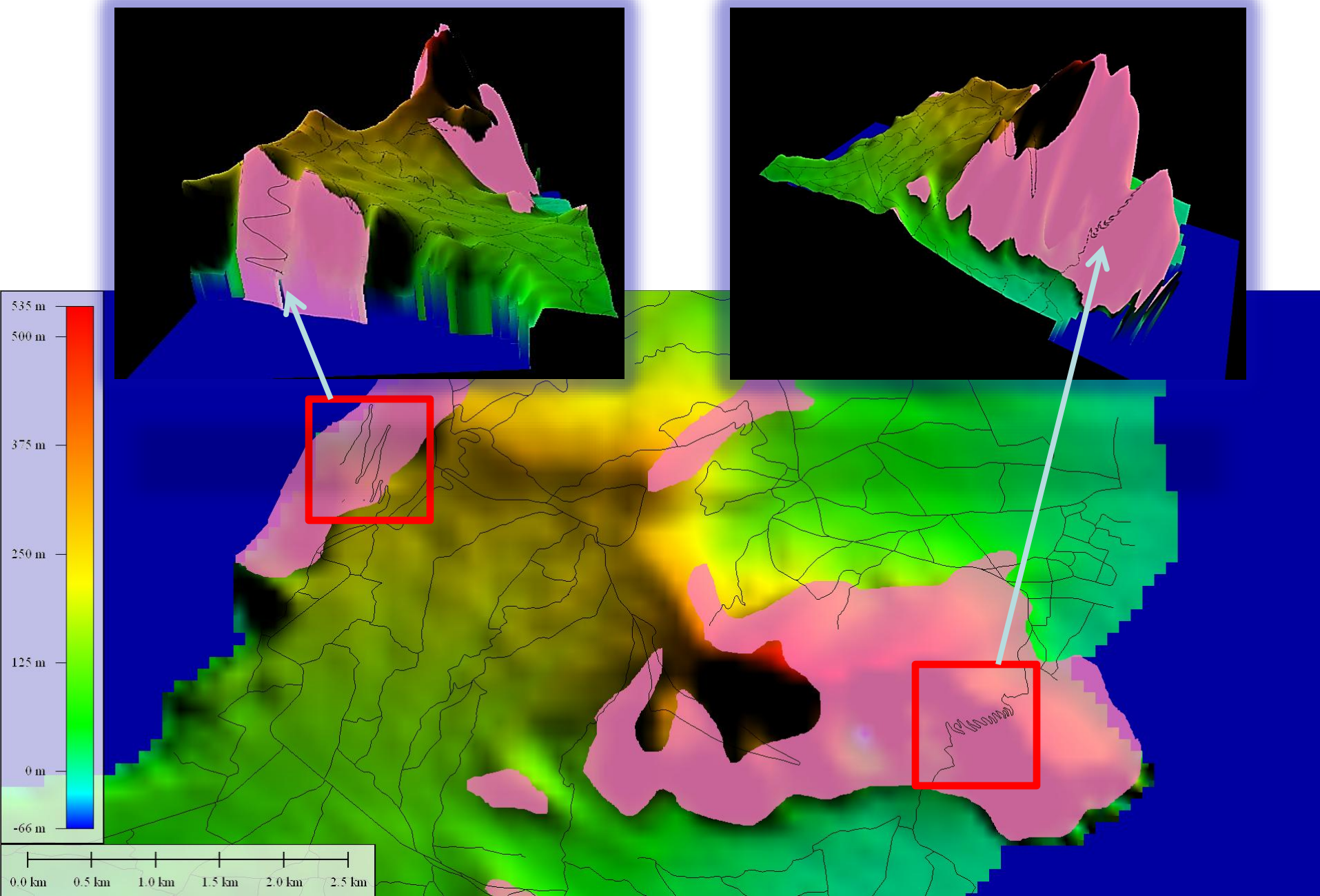
Access to outcrops (add Roads)



Zoom in and create a 3D map

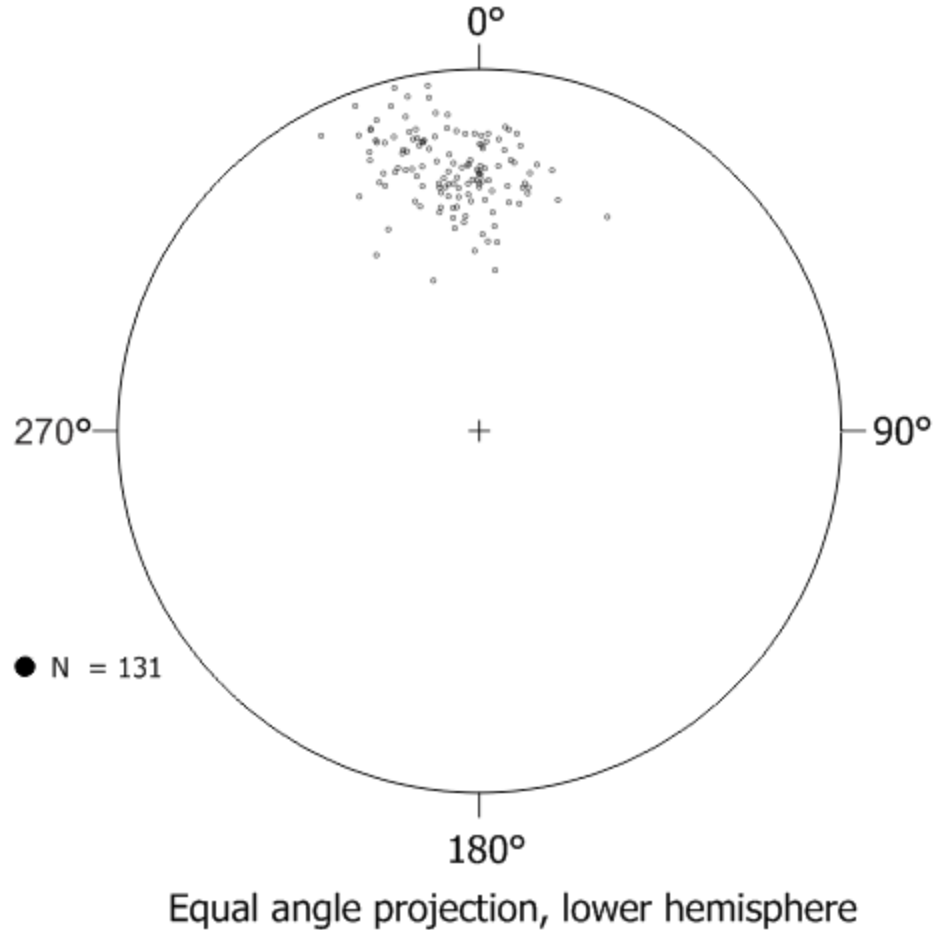


3D: access of rocks exposure

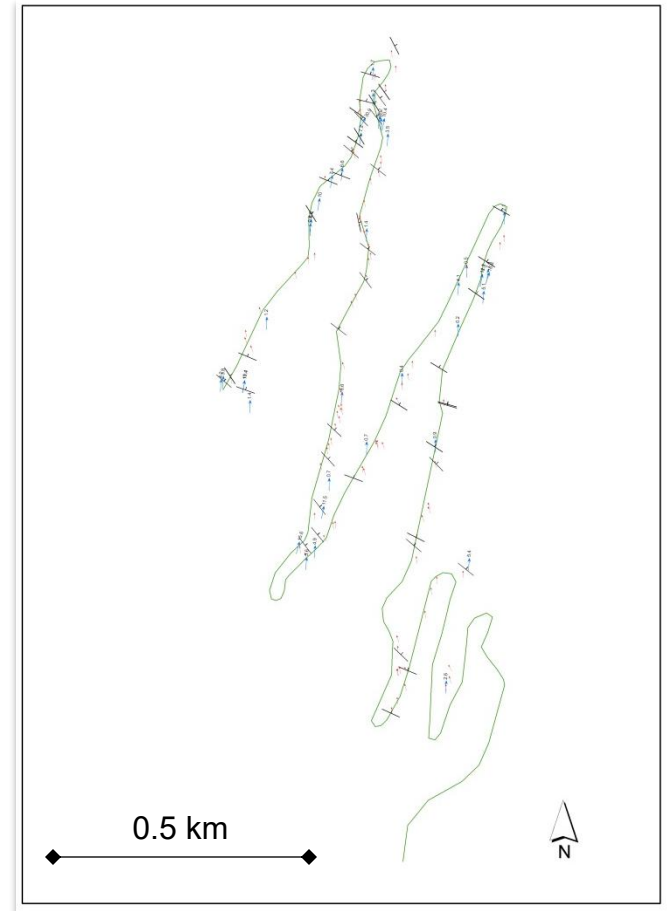


Raw data

- Stereonet (Lineation)

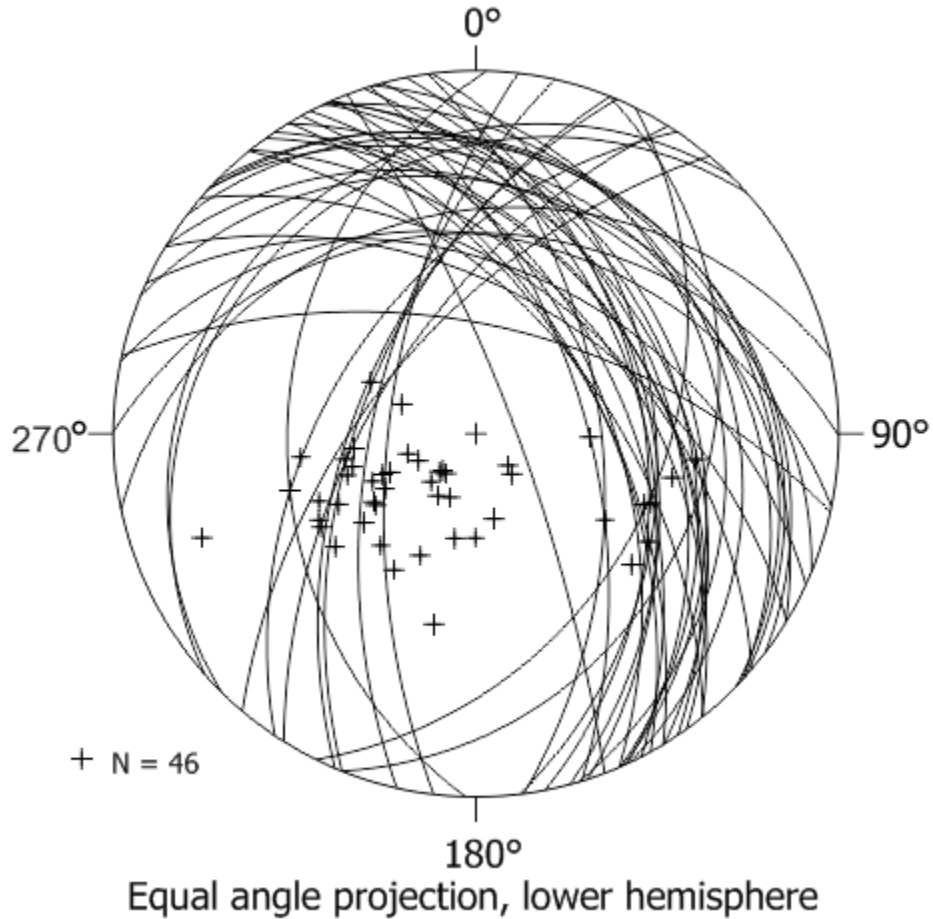


- Draft Map

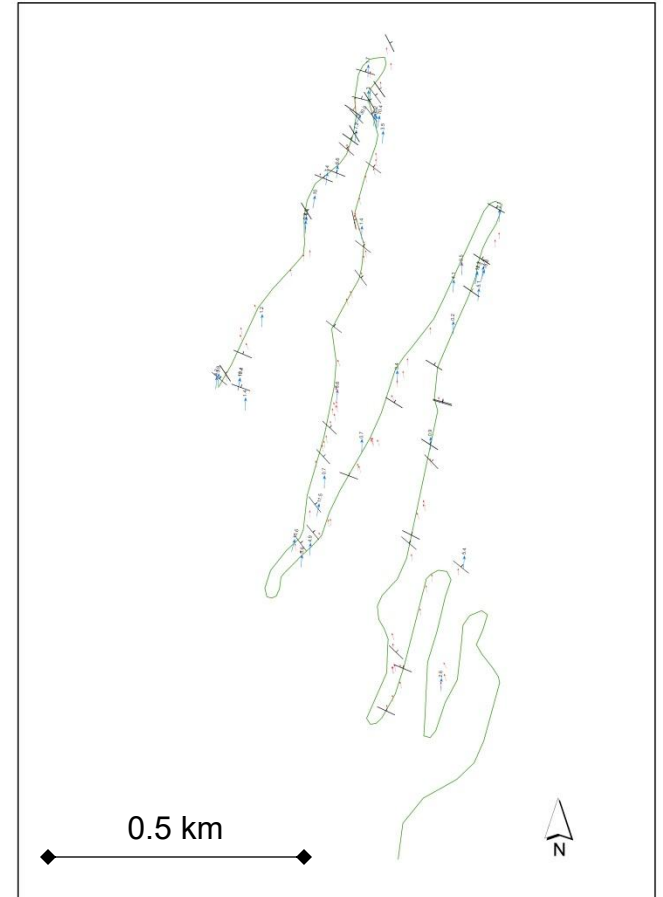


Raw data

- Stereonet (Foliation)

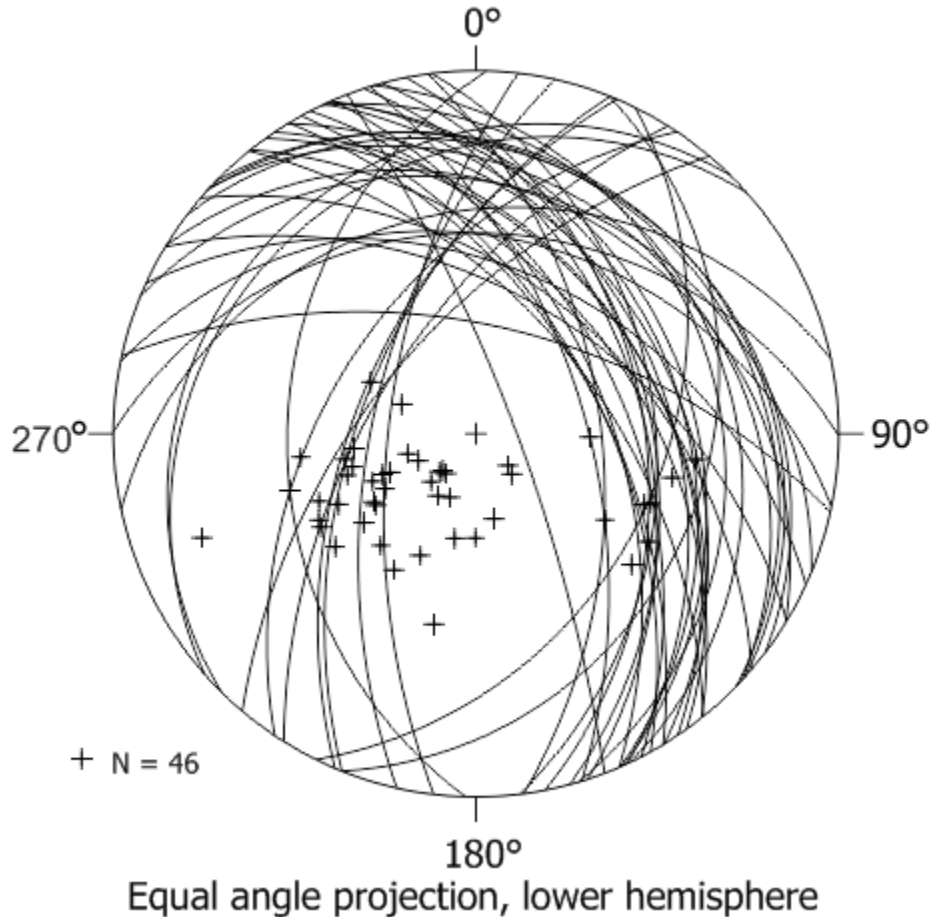


- Draft Map



Raw data

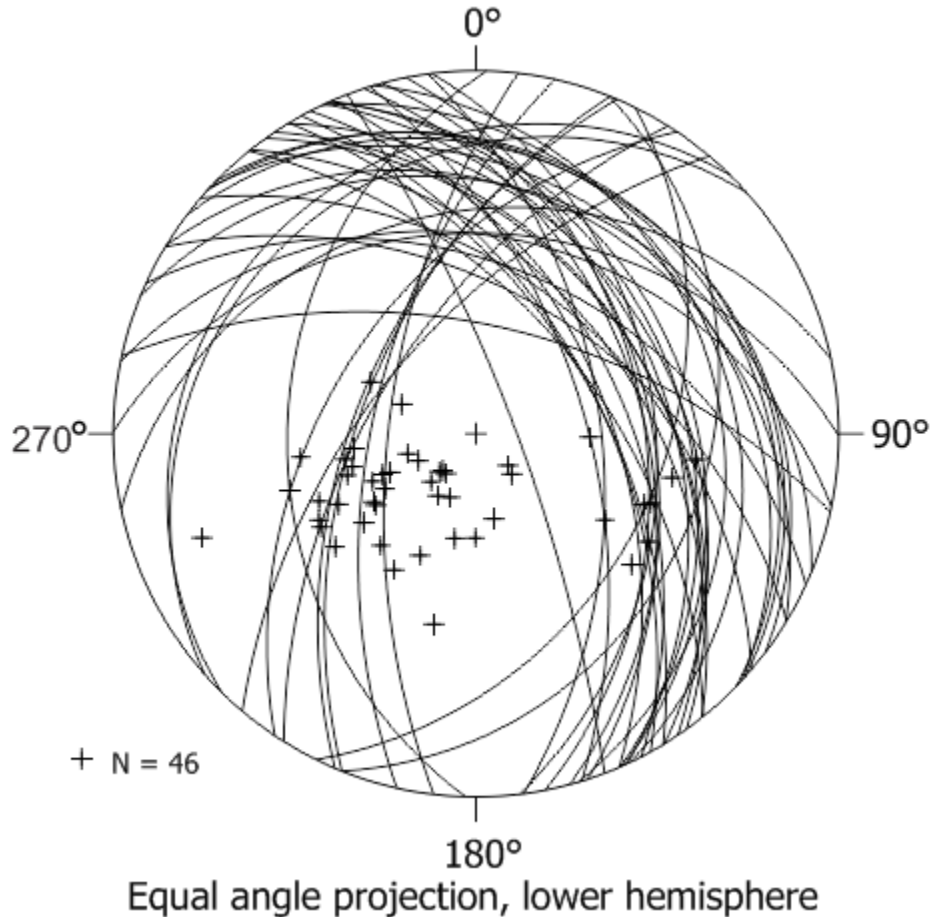
- Stereonet (Foliation)



- Shear sense
 - Top-N compressional (?)
 - Top-S compressional (?)
 - Top-S extensional

Raw data

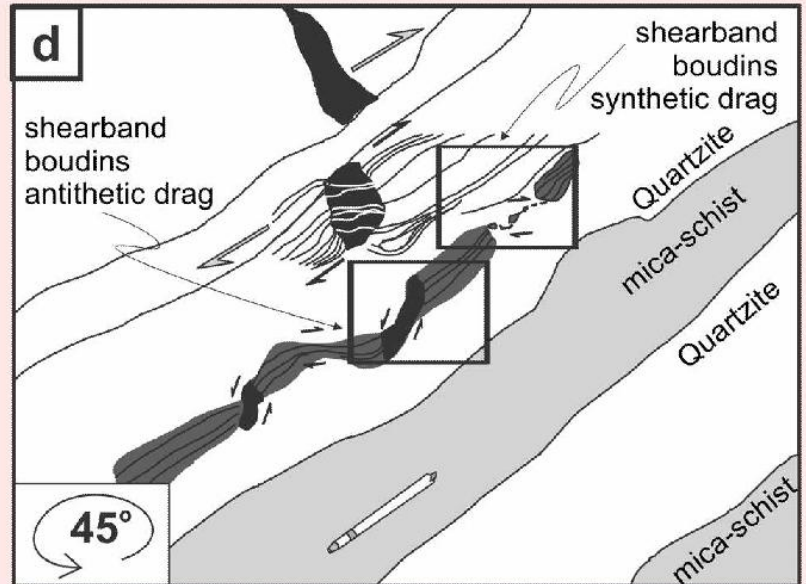
- Stereonet (Foliation)



- Shear sense

- Top-N **compressional** (?)
- Top-S **compressional** (?)
- Top-S extensional

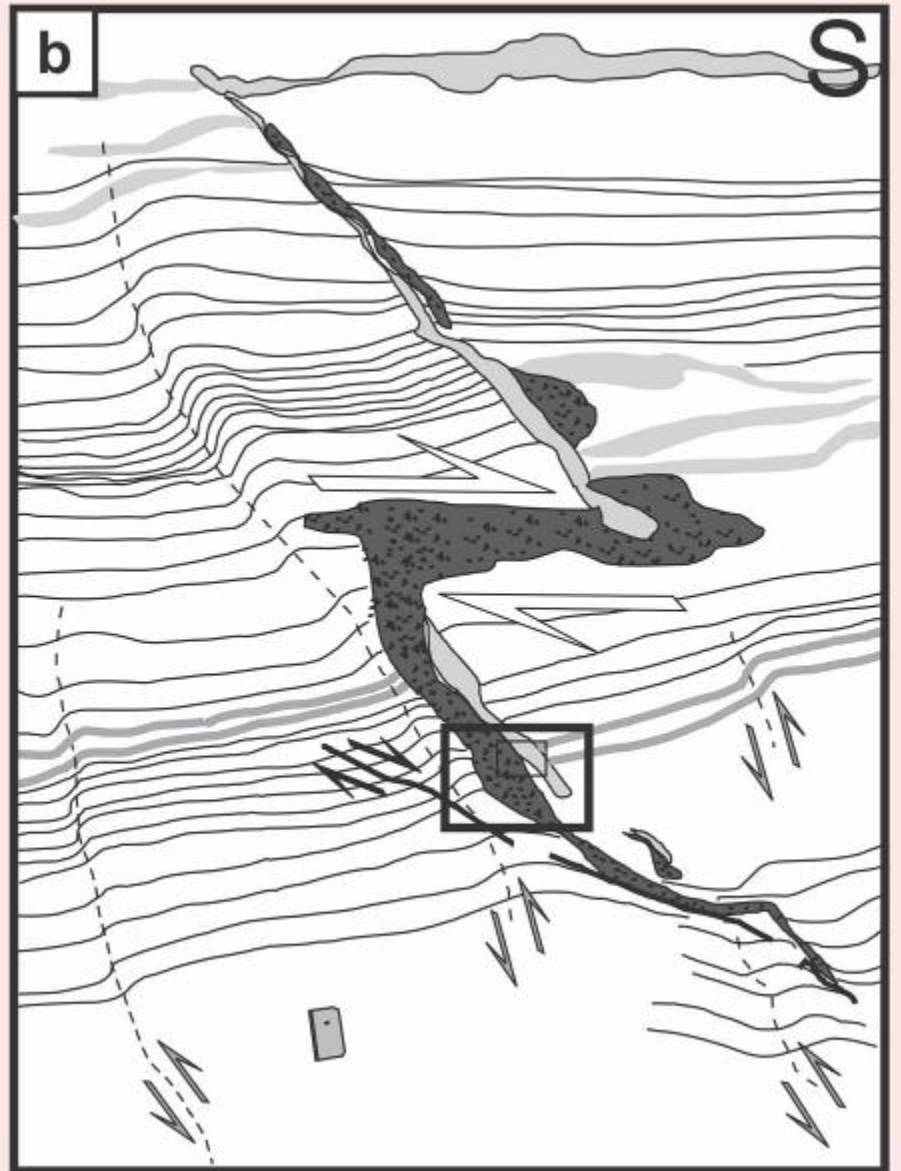
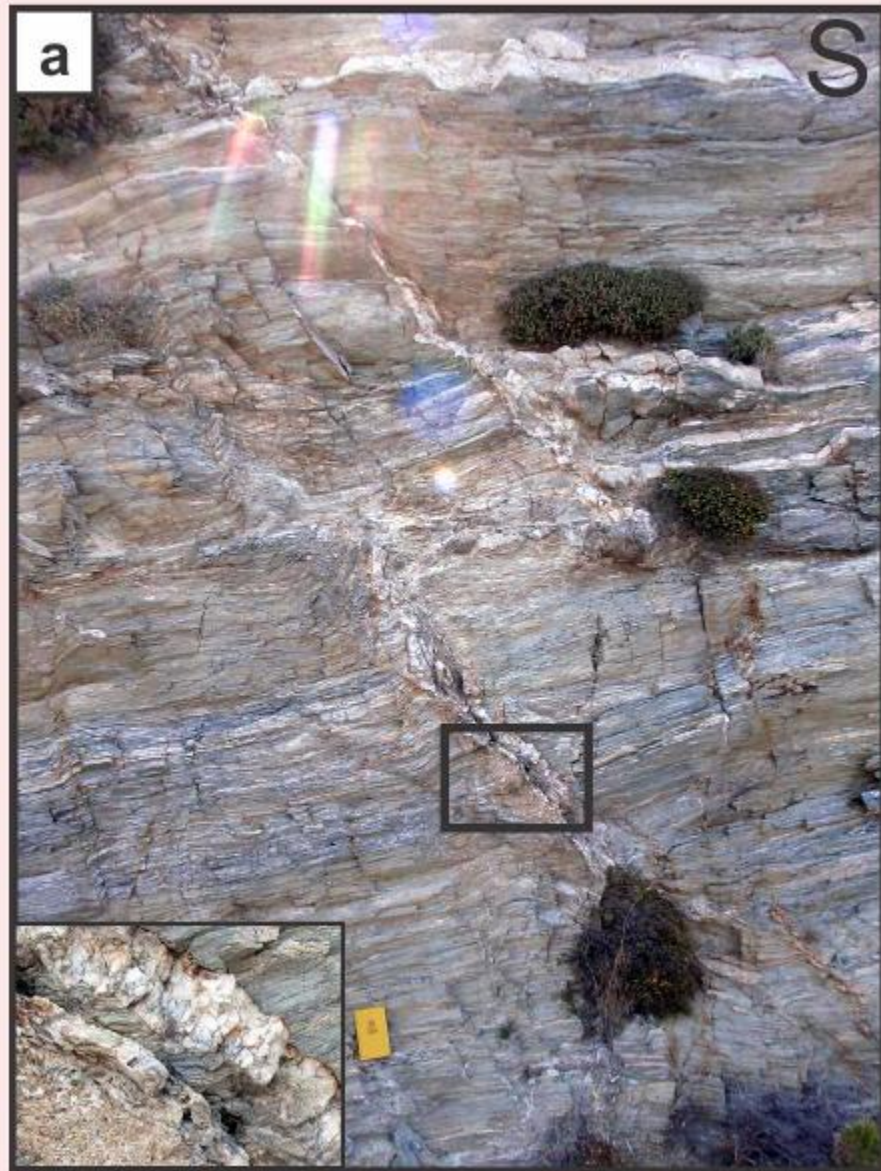
Structures that show Top-S (?)



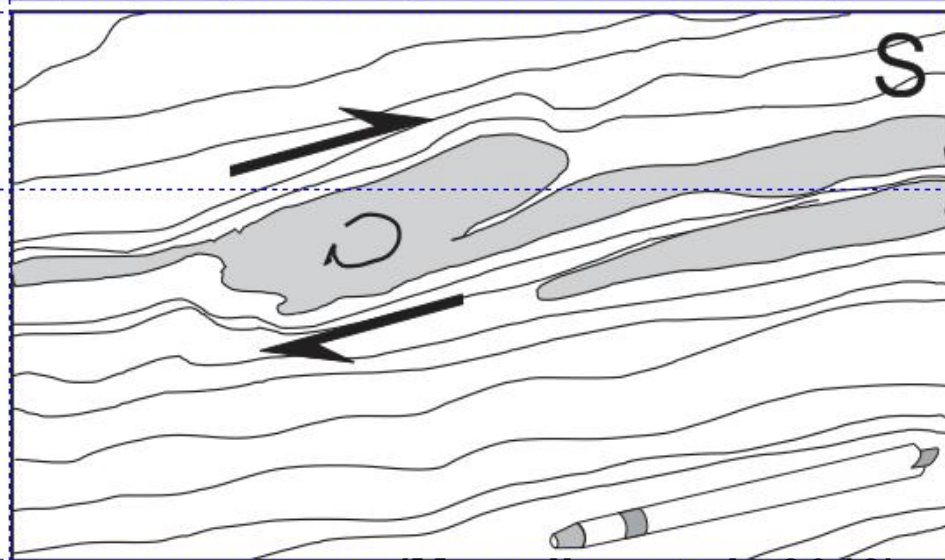
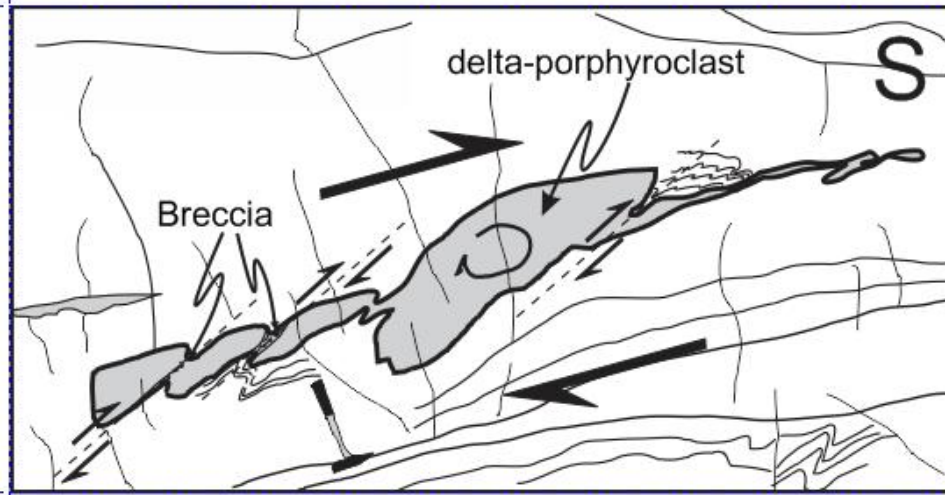
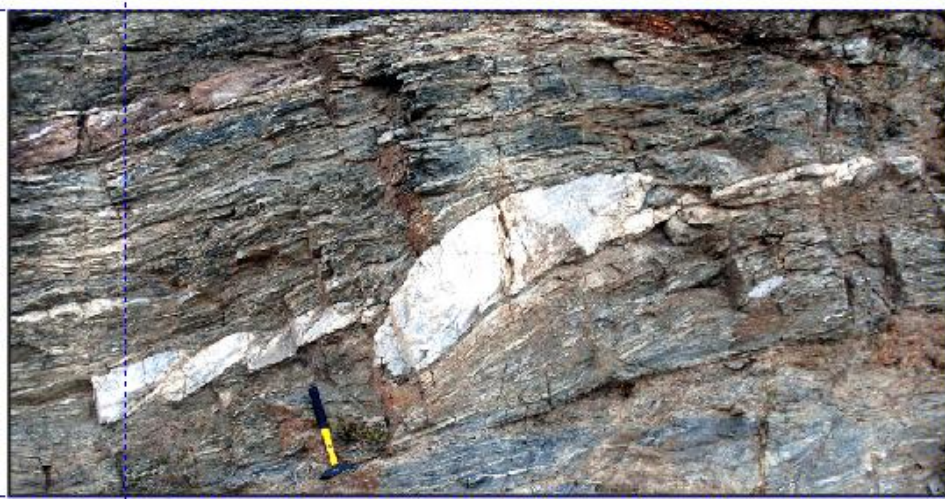
Structures show Top-S (?)



Structures show Top-S (?)

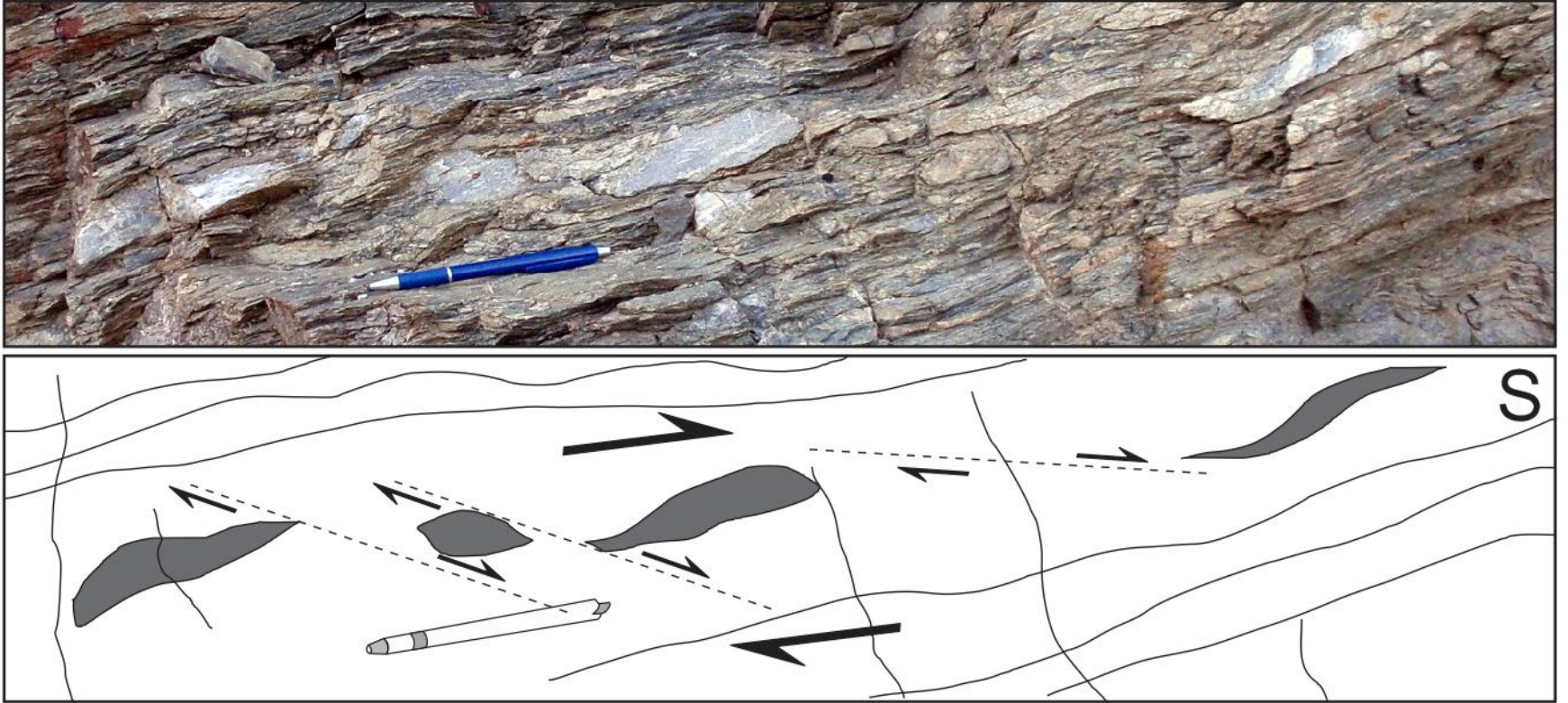


Structures show Top-S (?)

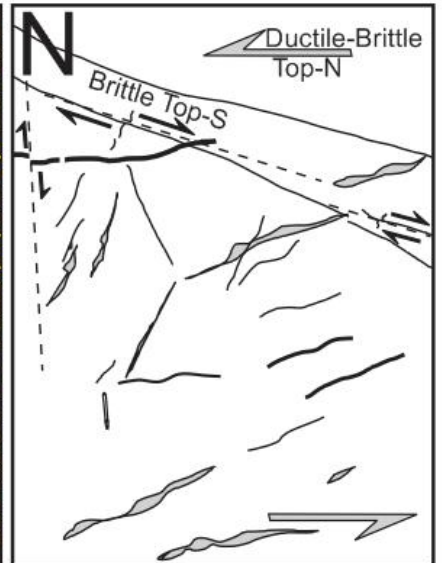
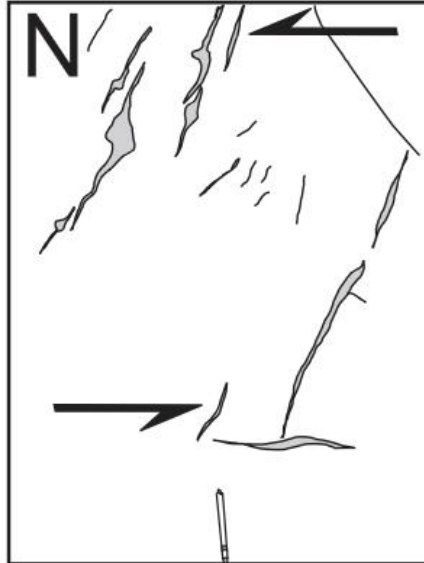


(Marsellos et al., 2012)

Structures show Top-S (?)

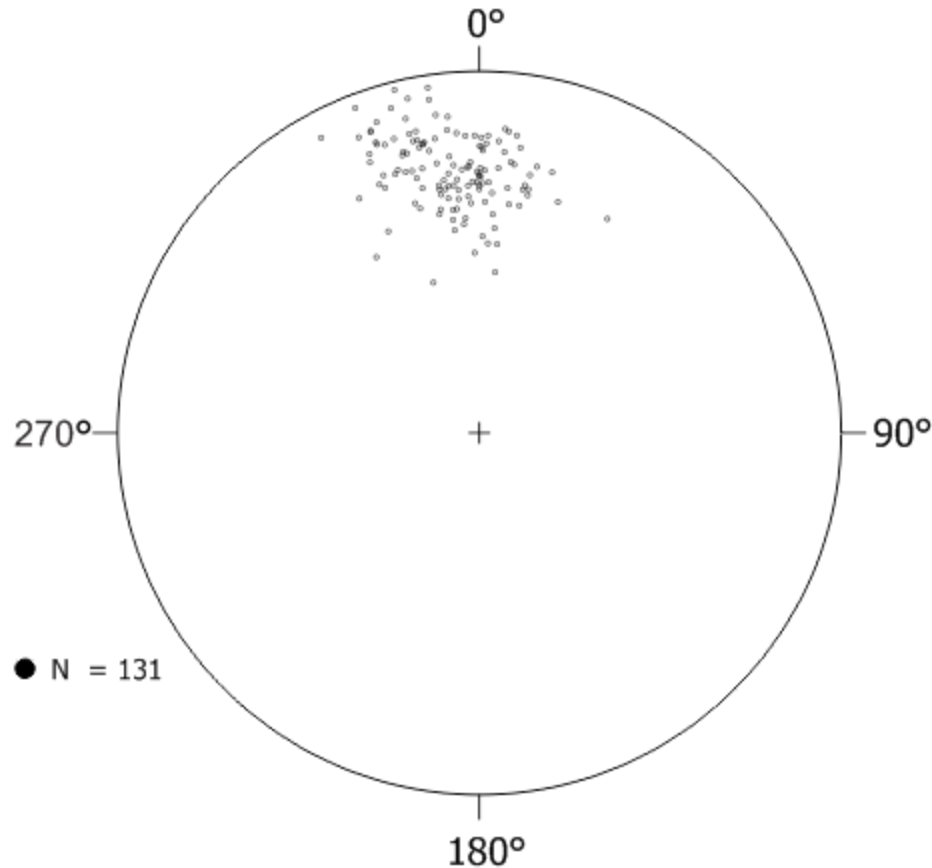


Structures show Top-N (?)



More than one metamorphic unit?

- Stereonet (Lineation)

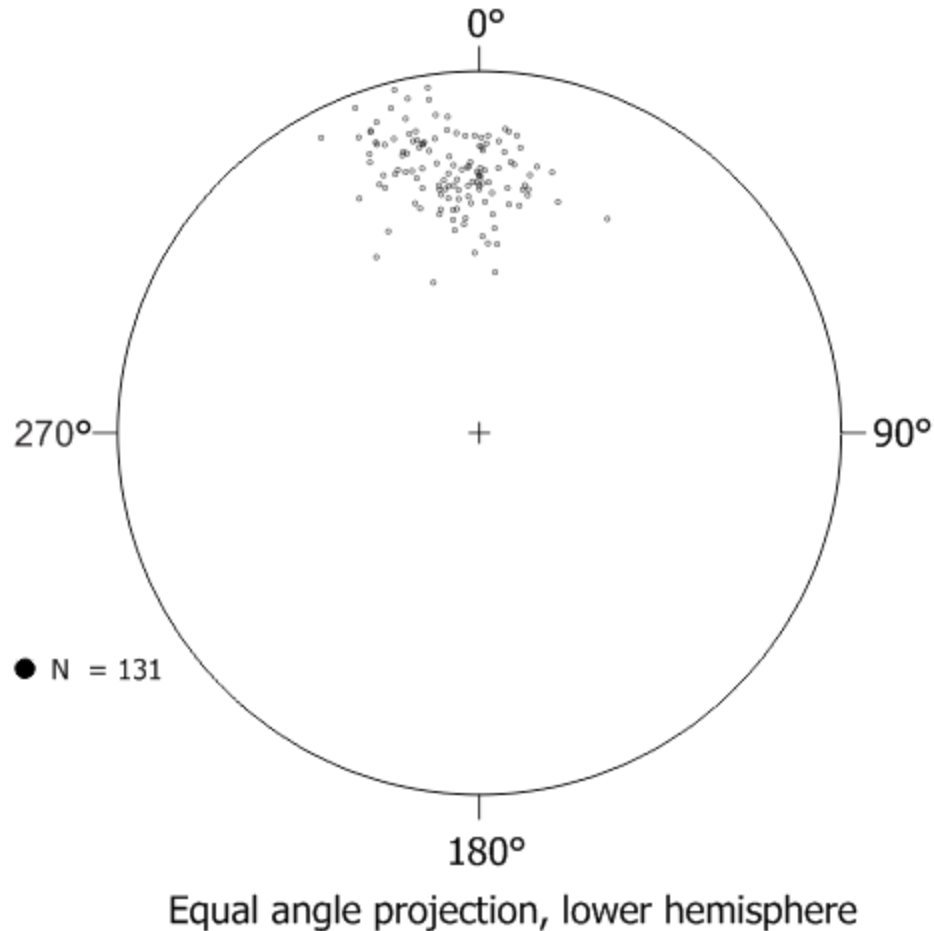


Equal angle projection, lower hemisphere

- Shear sense
 - *Brittle Data*
 - Top-S **thrust** faults (?)
 - *Ductile-Brittle Data*
 - Top-S & **Top-N**
 - *Ductile Data*
 - Top-S & **Top-N**

More than one metamorphic unit?

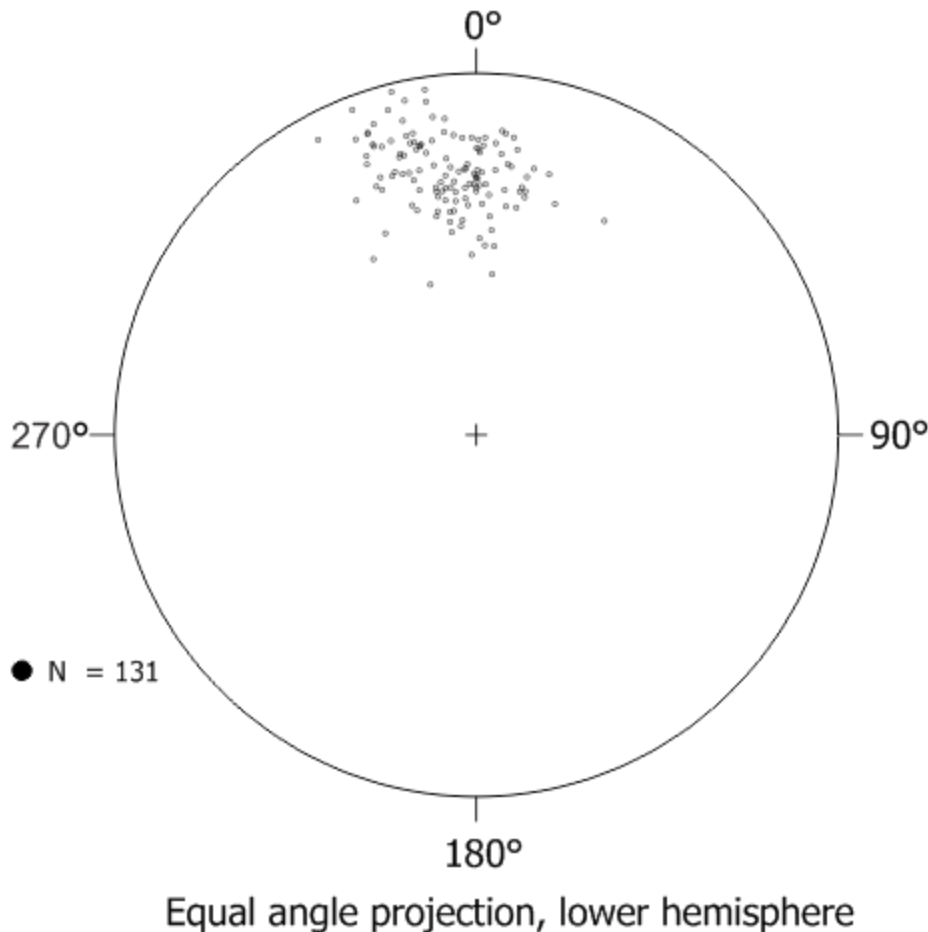
- Stereonet (Lineation)



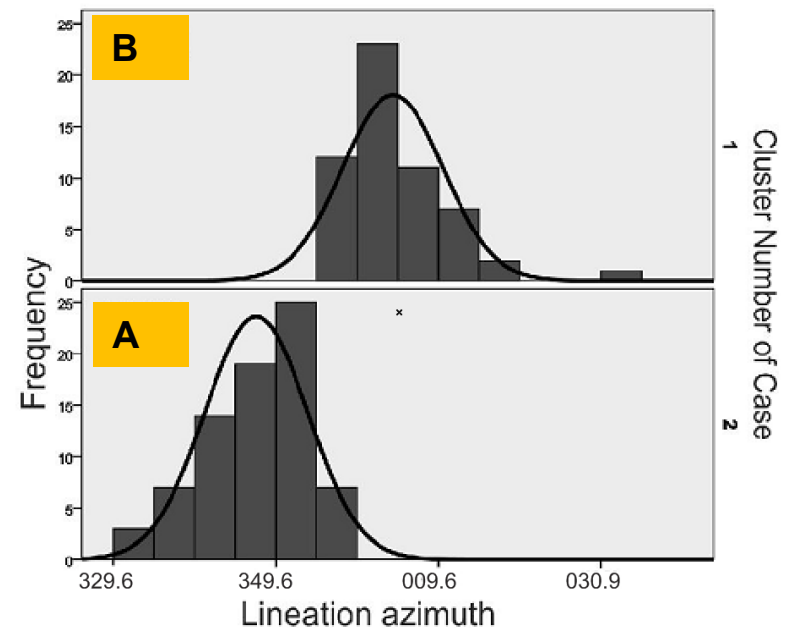
- [Cluster Analysis](#)

More than one metamorphic unit?

- Stereonet (Lineation)



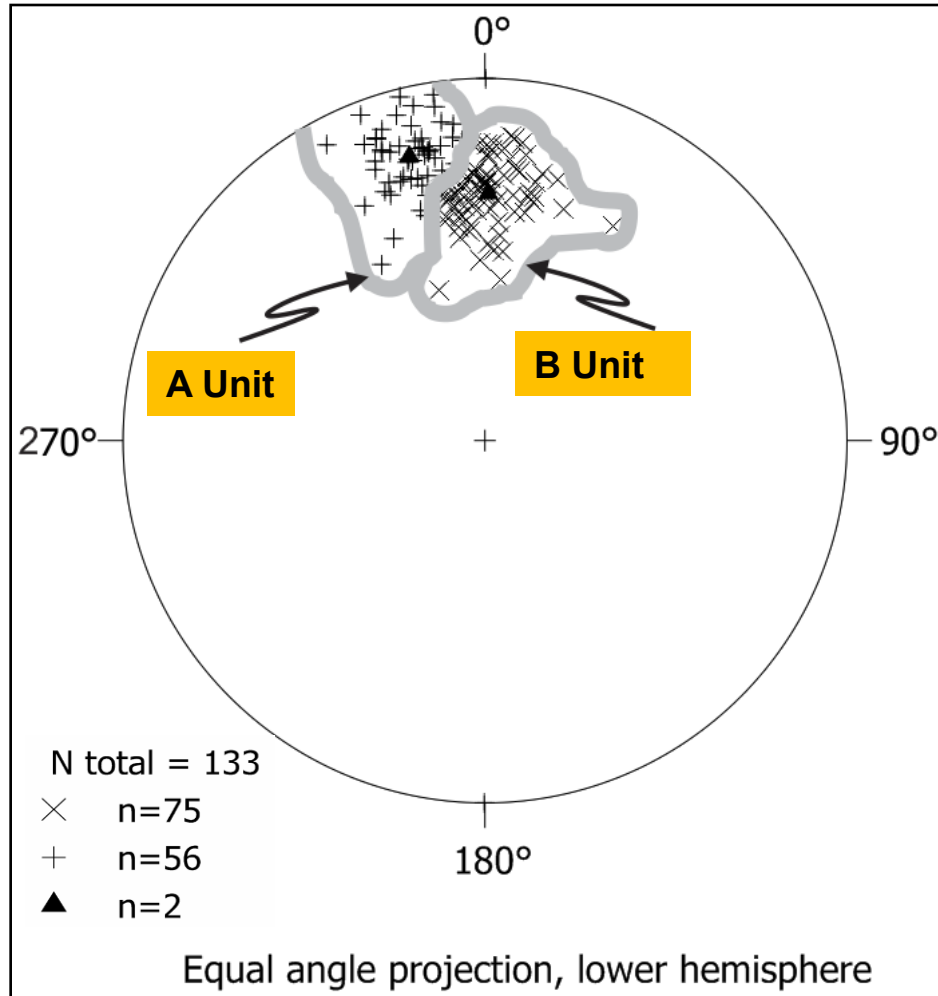
Cluster Analysis



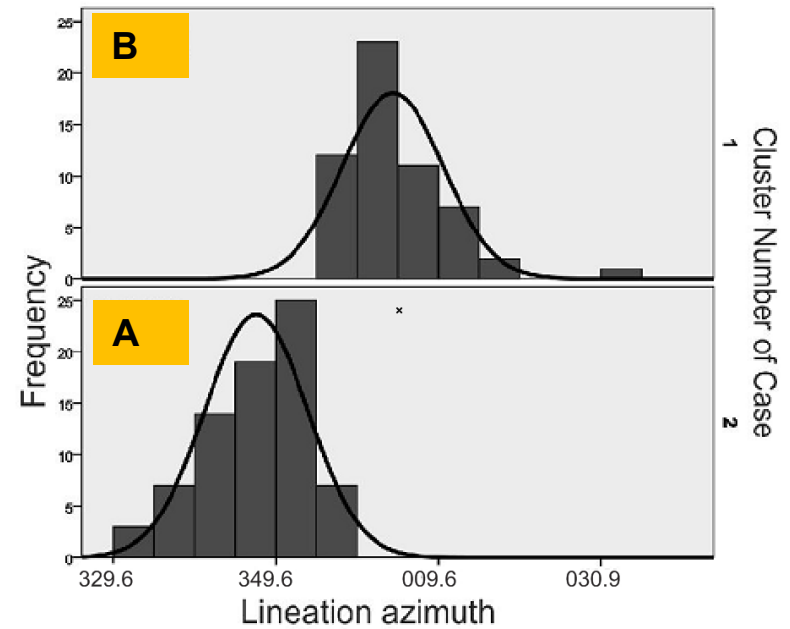
Rock Unit	N	Range	Minimum	Maximum	Mean	Std. Error	Std. Deviation
B Unit	56	33.60	357.30	30.90	3.97	0.83	6.19
A Unit	75	26.70	329.60	356.30	347.17	0.73	6.34

Two clusters = two metamorphic units

- Stereonet (Lineation)



Cluster Analysis

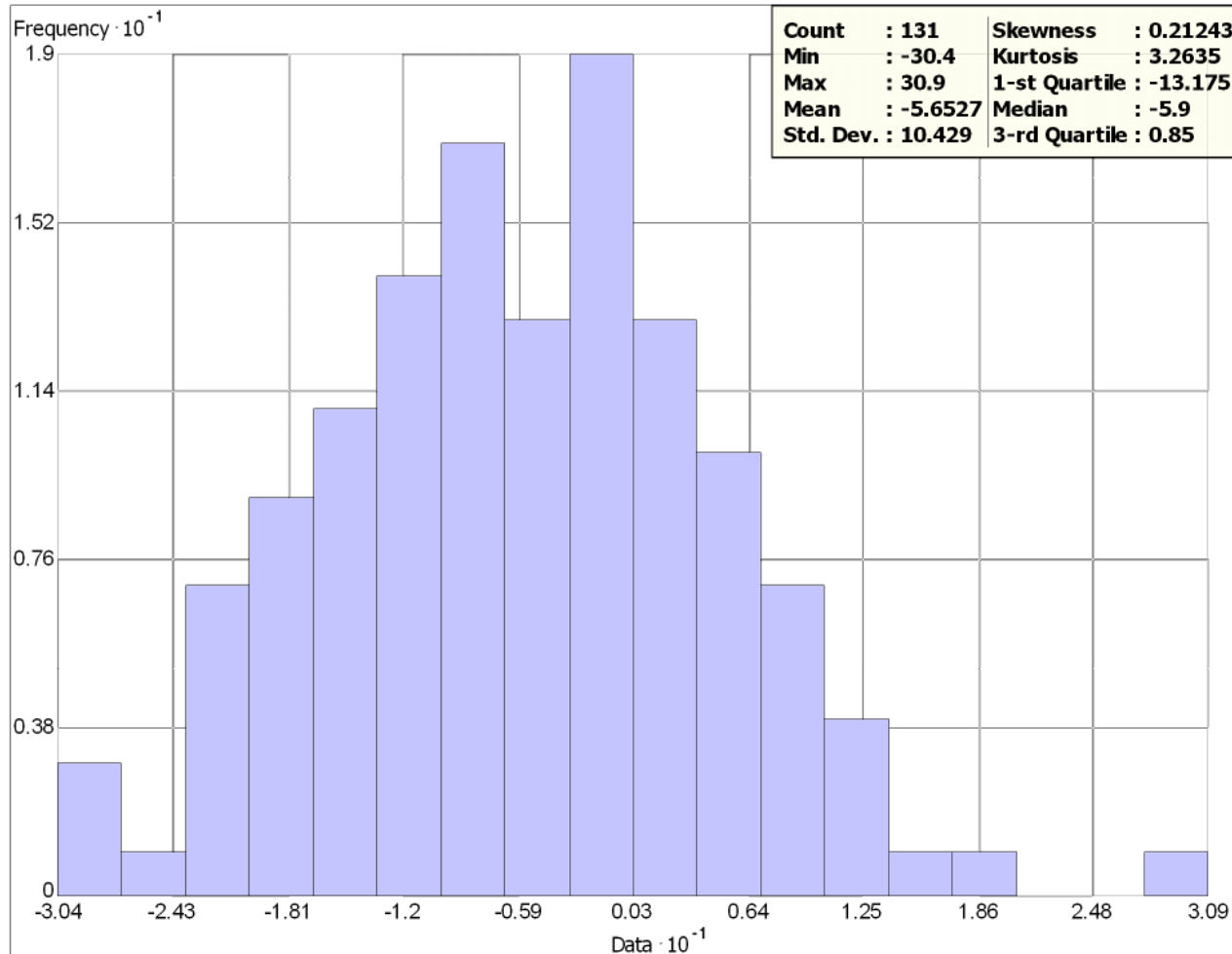


Rock Unit	N	Range	Minimum	Maximum	Mean	Std. Error	Std. Deviation
B Unit	56	33.60	357.30	30.90	3.97	0.83	6.19
A Unit	75	26.70	329.60	356.30	347.17	0.73	6.34

Exploring the data

- 1. Do the lineation data show a normal distribution?

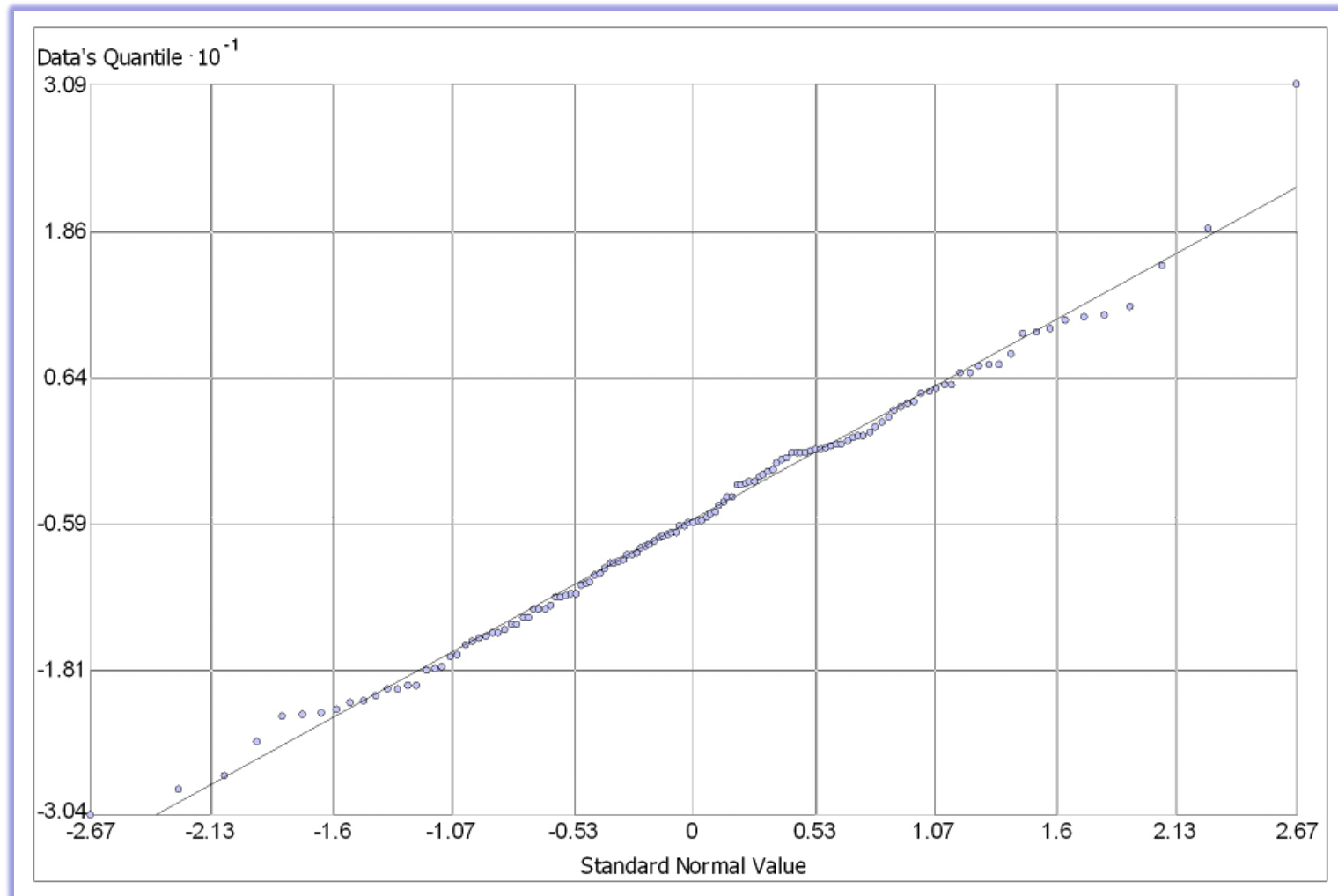
Histogram



Exploring the data

- 2. Data follow a normal distribution (Q-Q plot)

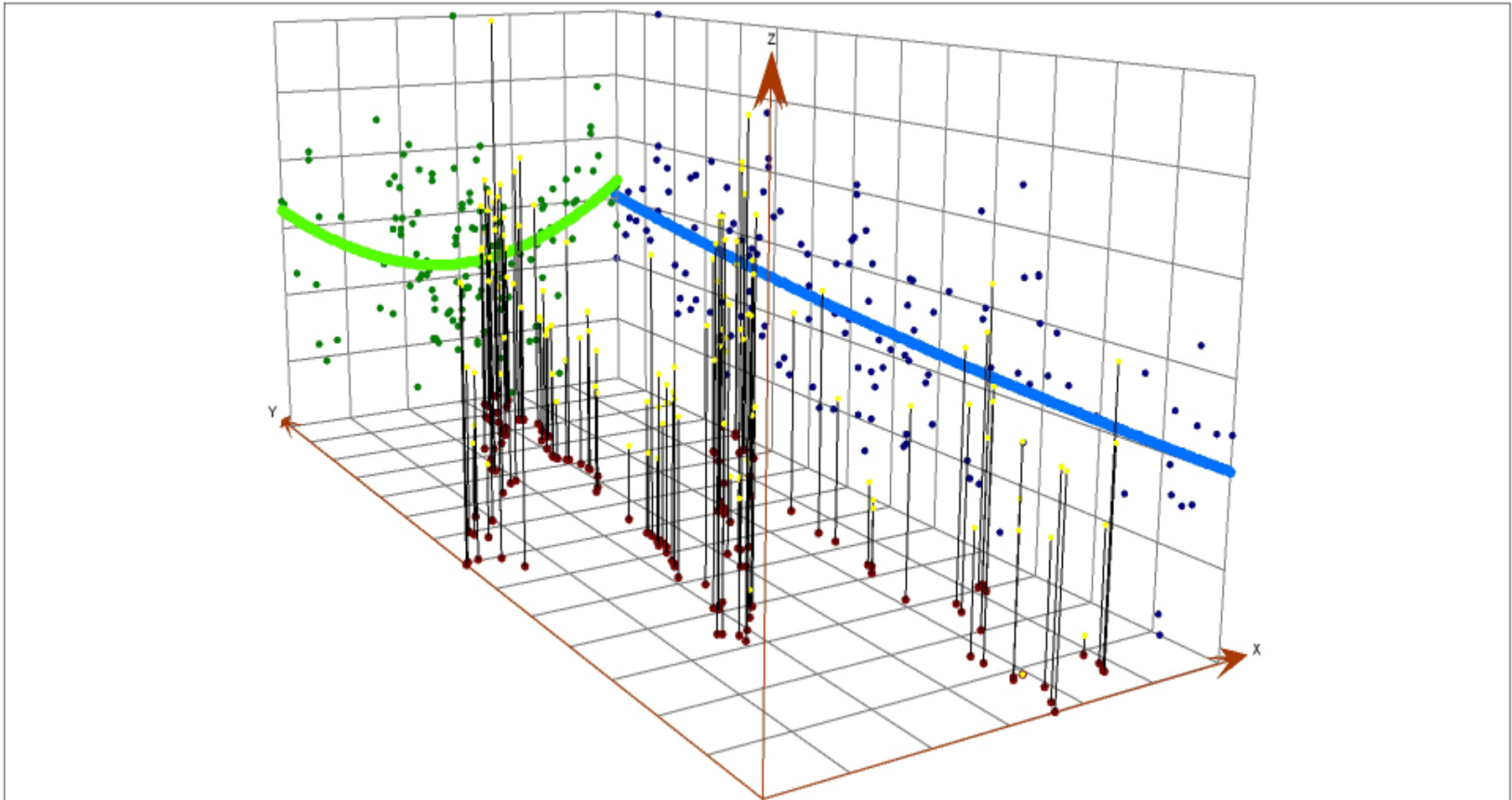
Normal Q-Q Plot



Extract patterns and fit a model

- 3. Are there any significant trends on the lineation data

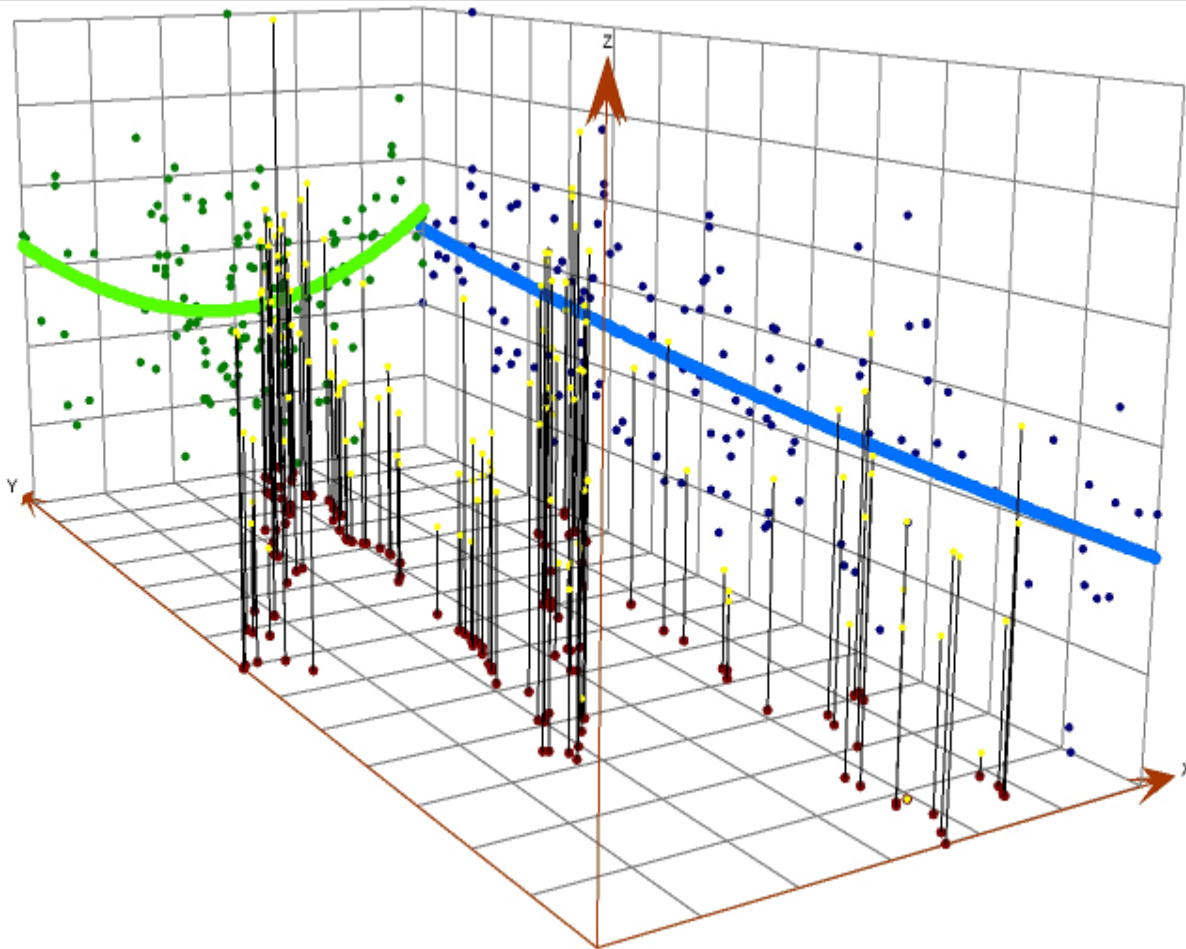
Trend Analysis



Extract patterns and fit a model

- 3. YES, a 2nd order polynomial trend (green)

Trend Analysis

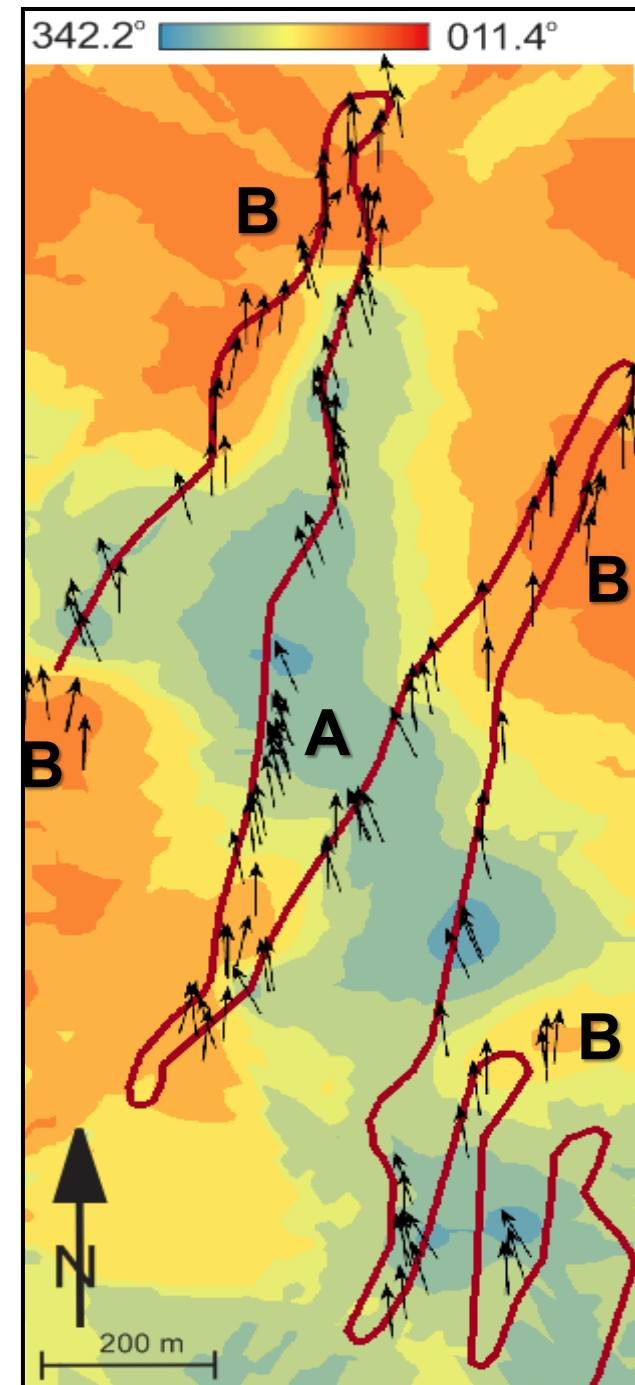


Create a prediction map

- 4. Ready to produce the prediction map of the two metamorphic units (A,B) exposure using **ordinary kriging**

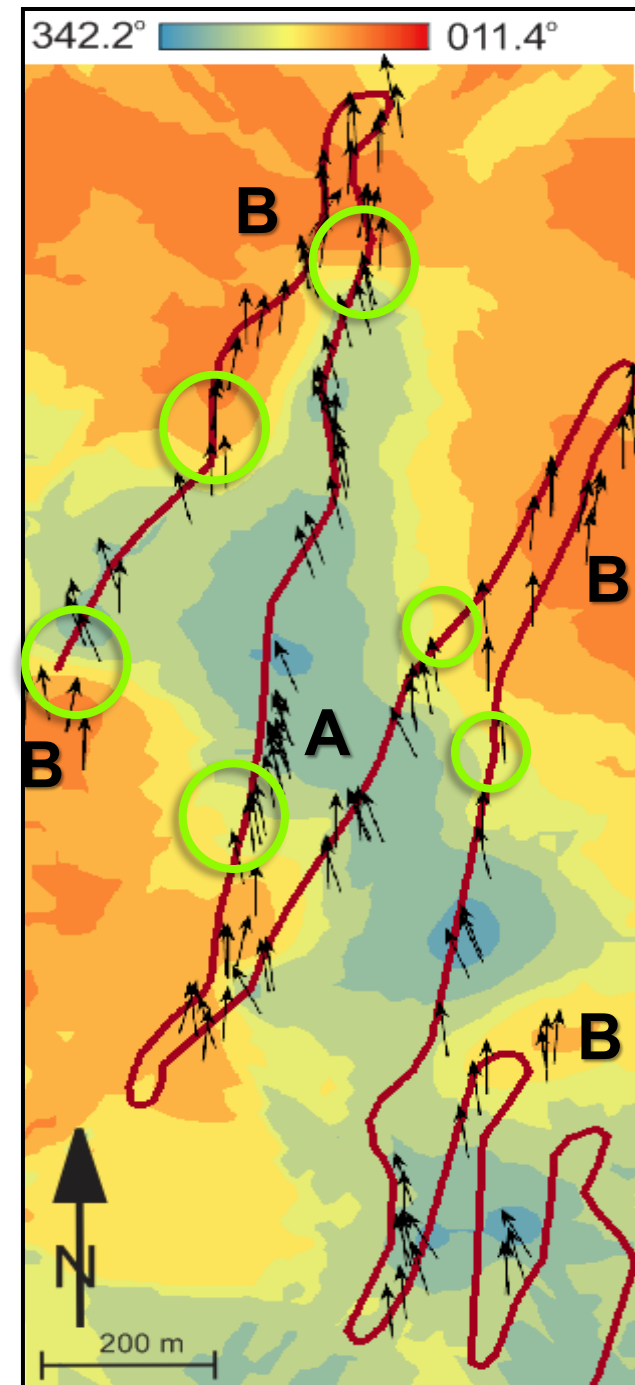
Ordinary kriging:

An interpolation method that depends on the normal distribution of the data.



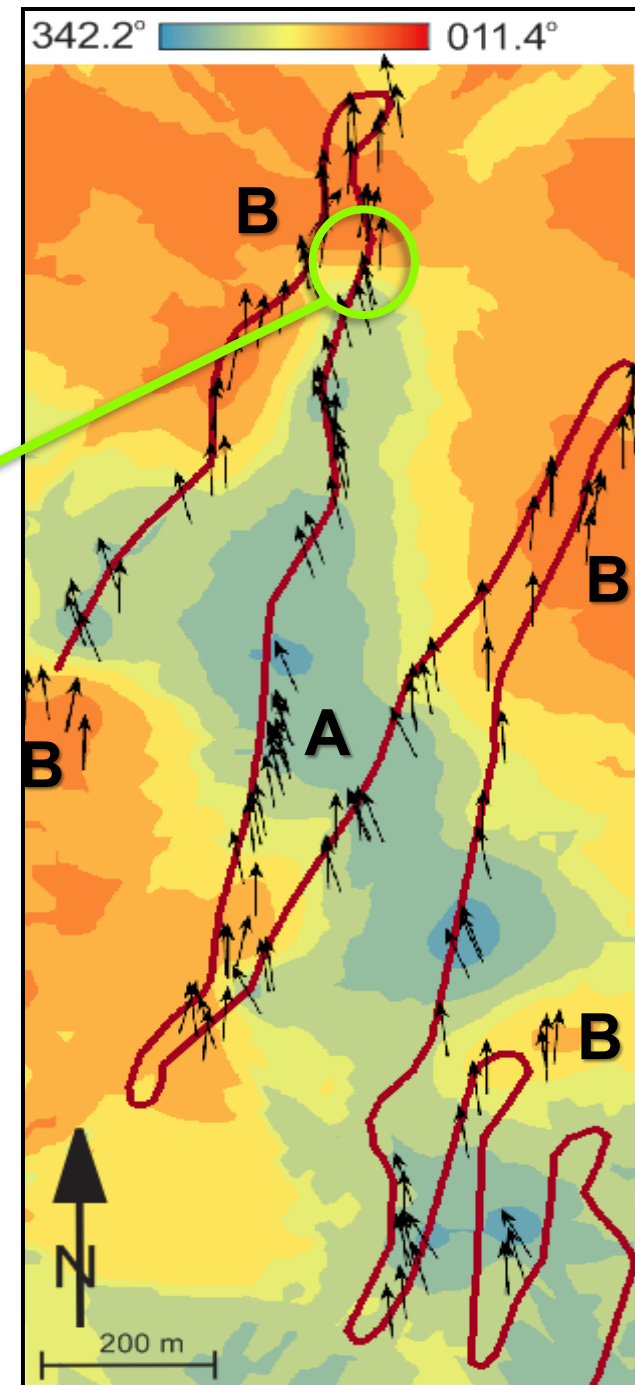
Create a prediction map and find the sampling locations

- 5. Best sampling locations for dating, petrological and micro-structural analysis



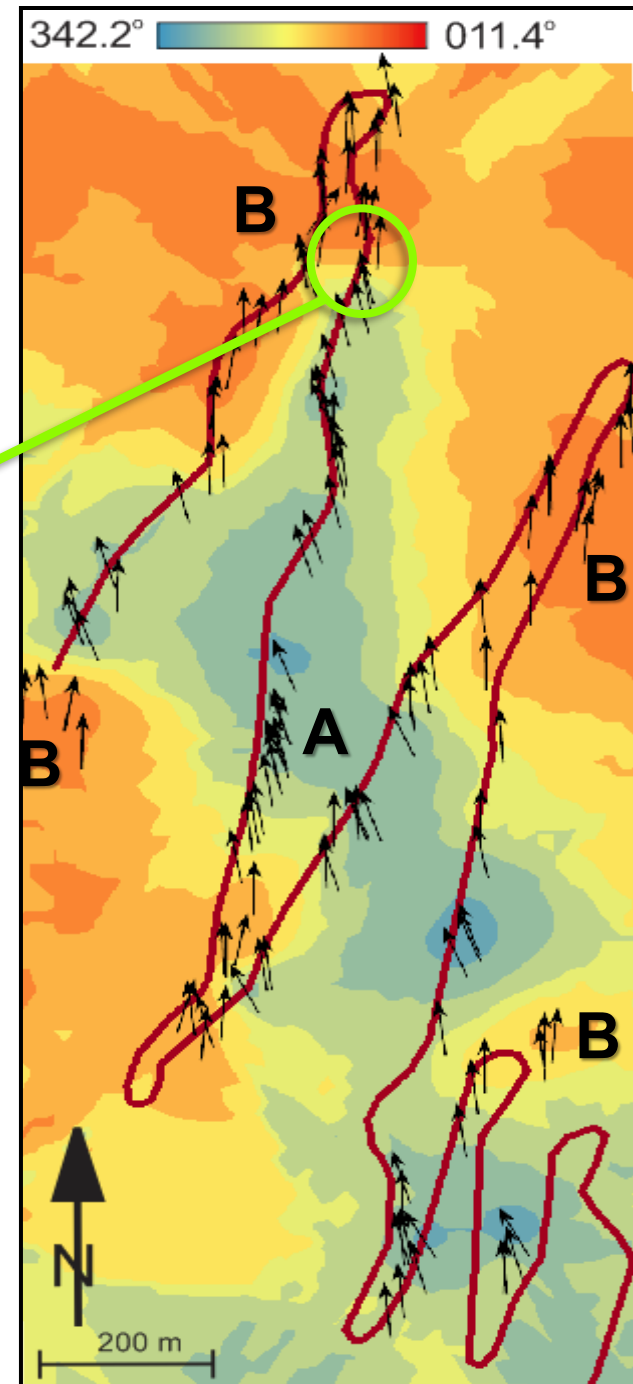
Sampling

- 6. Locate possible contact locations



Results

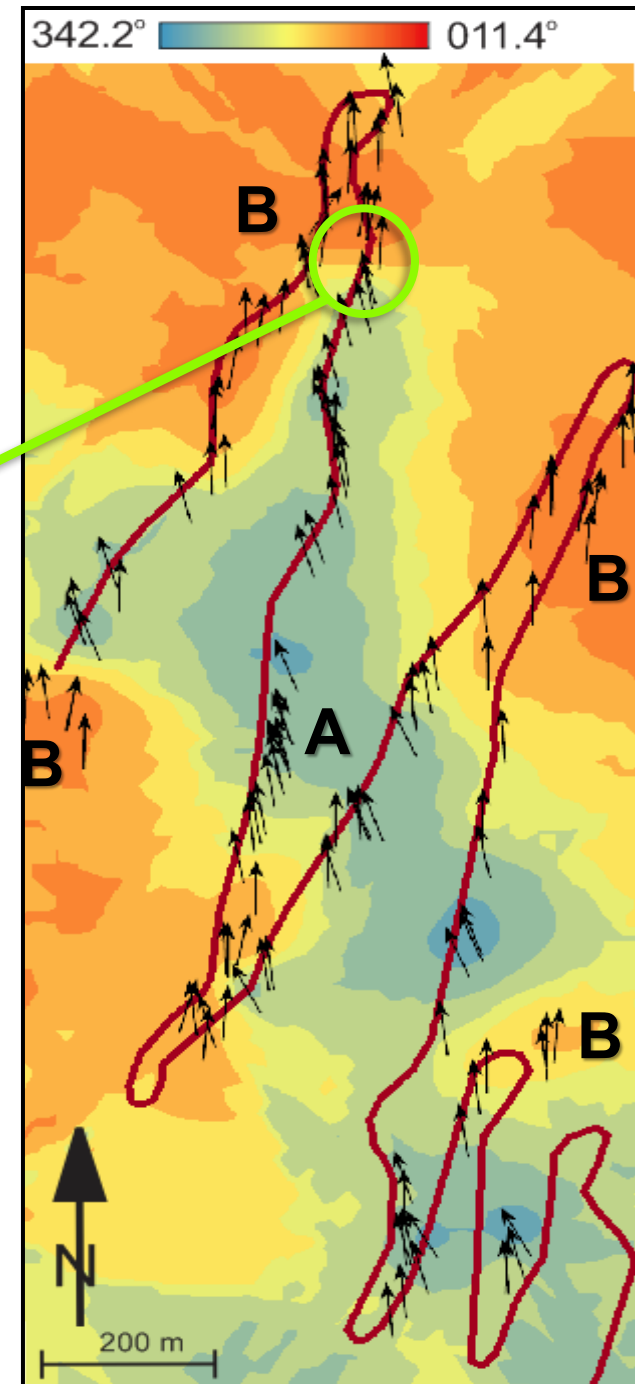
- 6. Dating results: Normal Fault(s)?



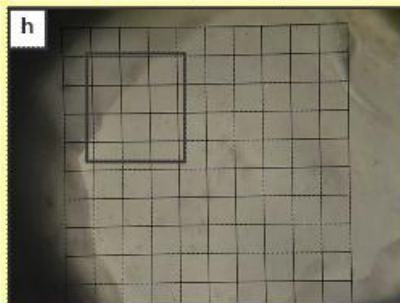
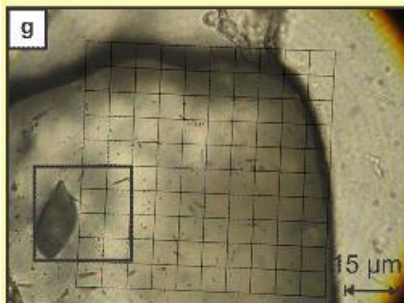
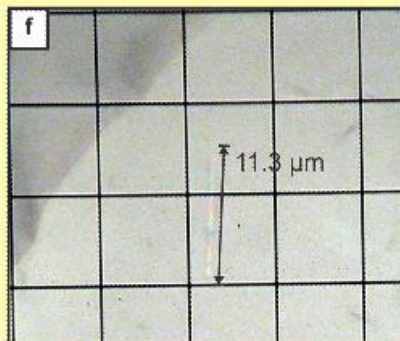
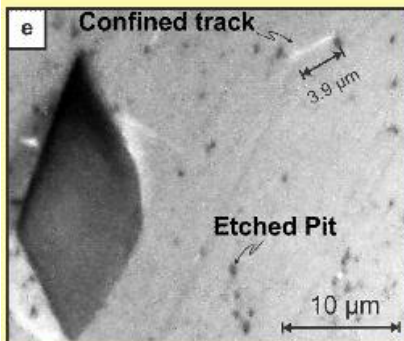
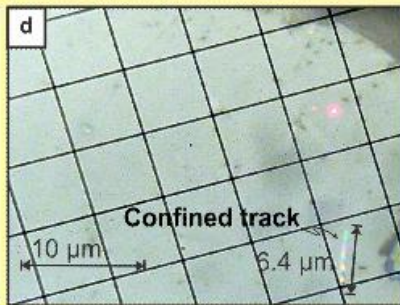
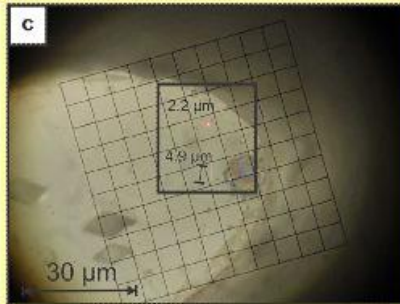
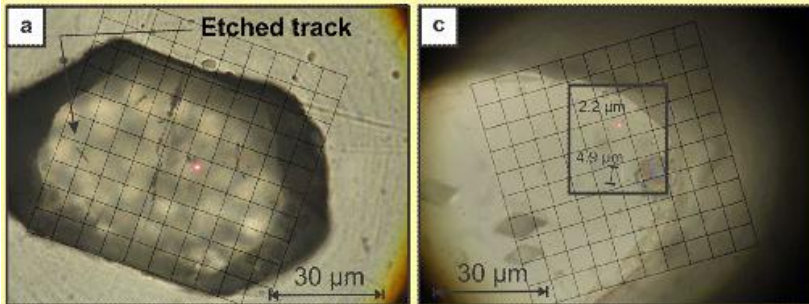
(Marsellos et al., 2013)

Results

- 6. Dating results

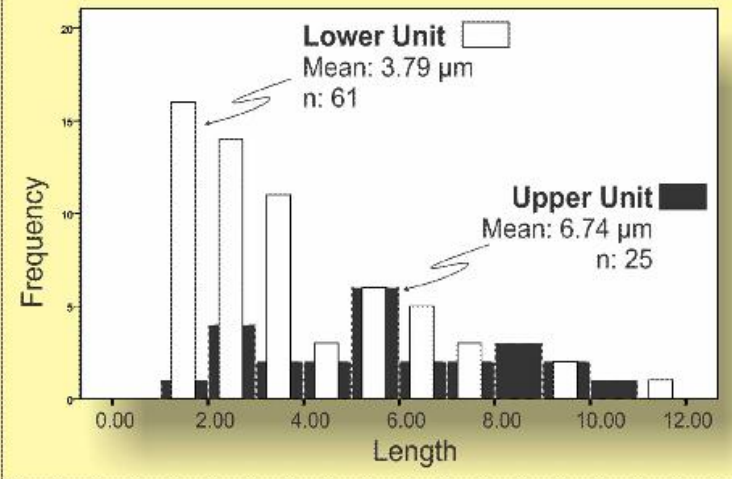


A younger below an older met. unit



Unreset zircons
from Upper unit (~ 48 Ma)

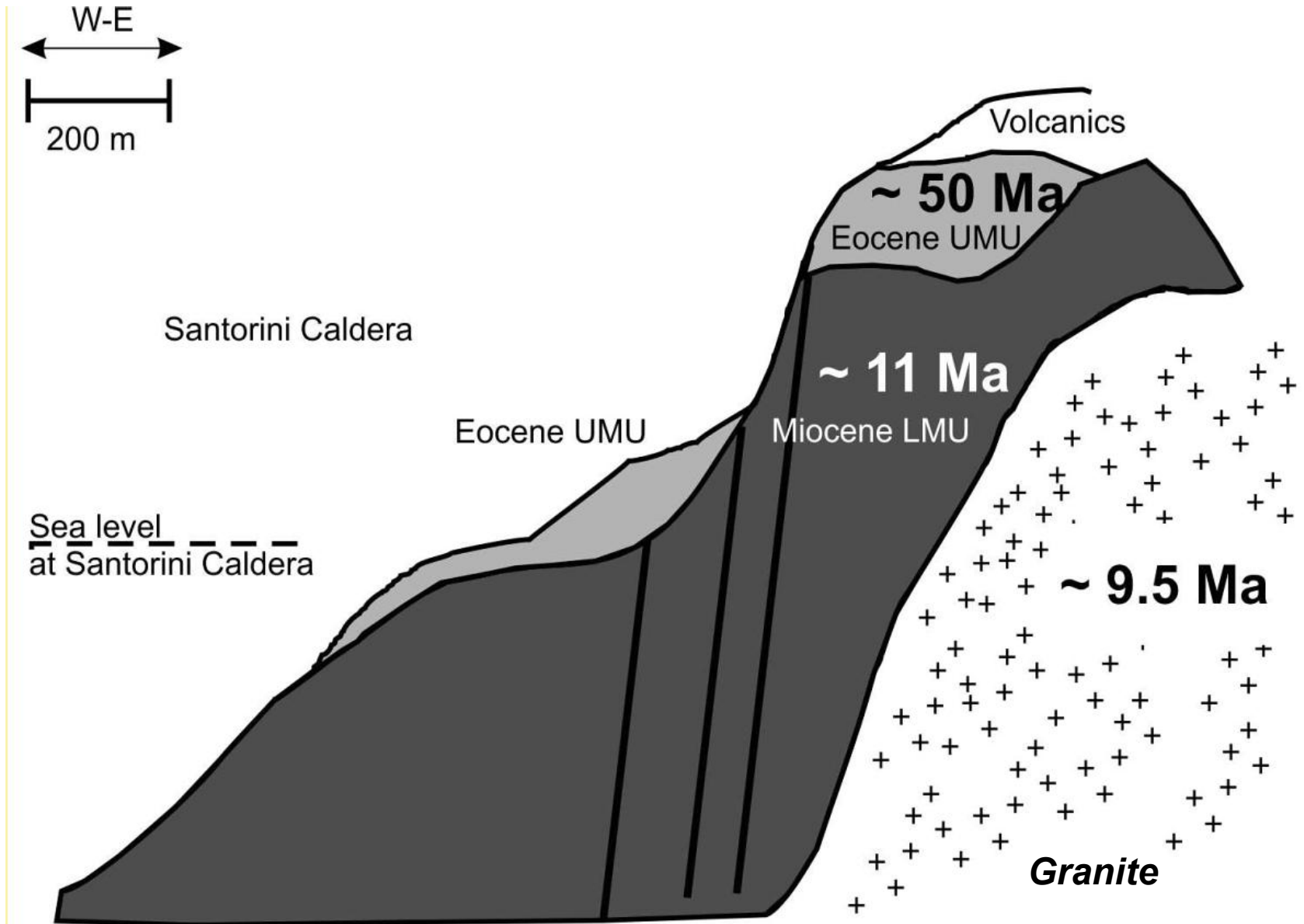
Zircon Fission-Track Length measurements
from Upper and Lower units



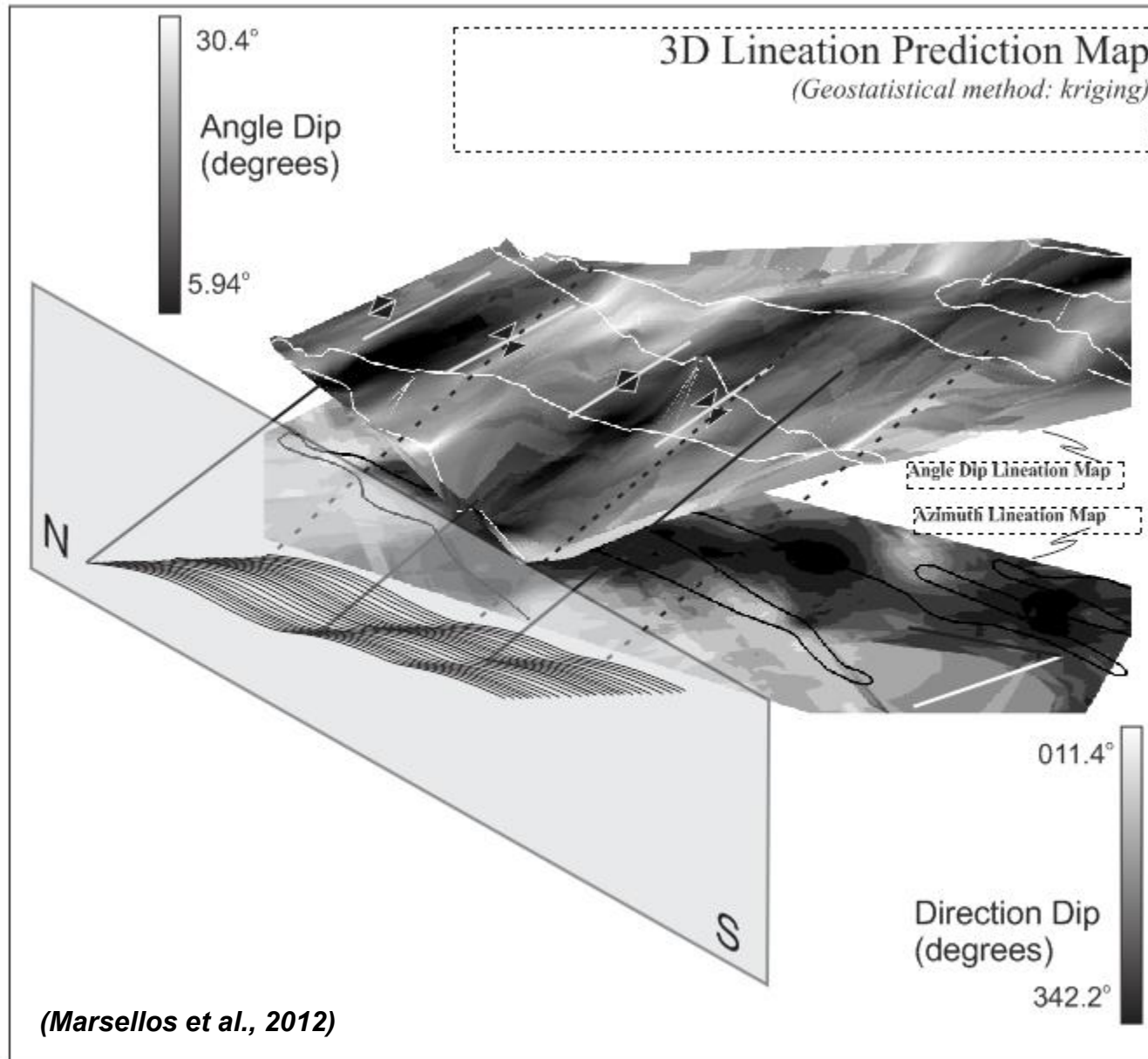
Reset zircons
from Lower unit (~ 11.5 Ma)

(Marsellos et al., 2012)

A younger below an older met. unit

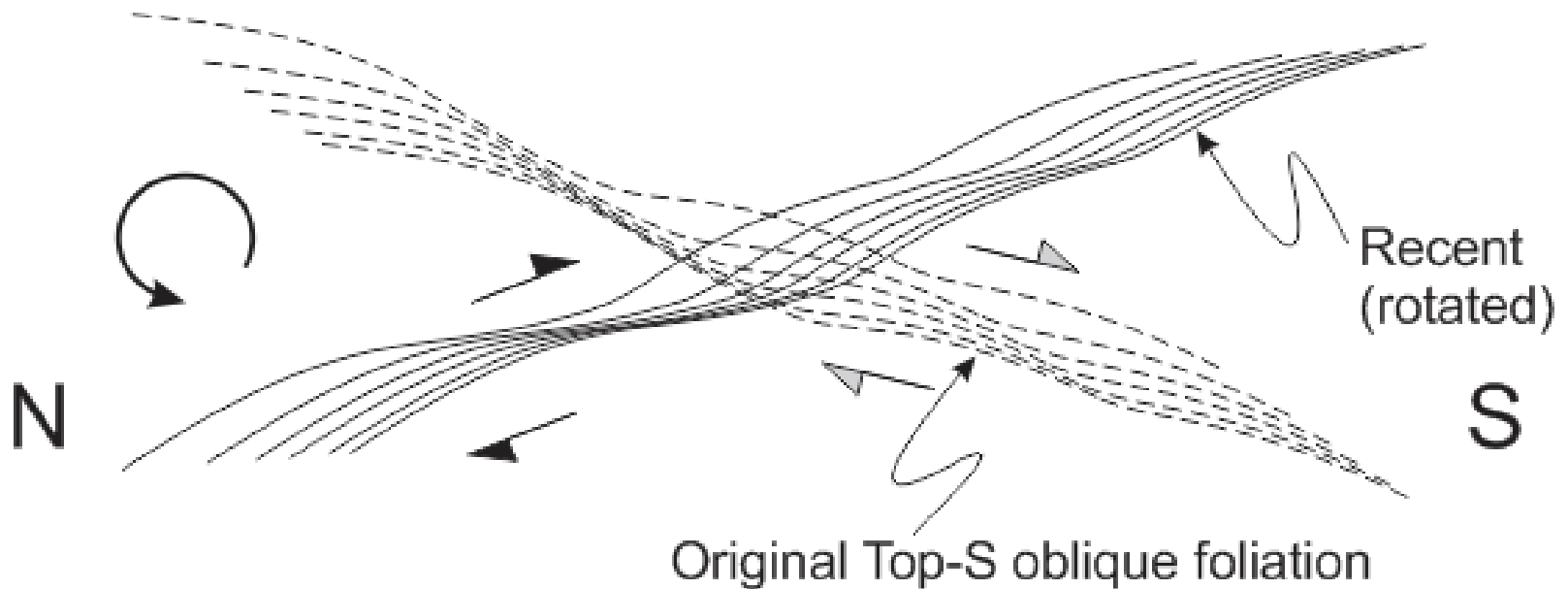


3D of all lineation dip angles



A mega-scale S/C structure (rotated)

Top-S
Rotated oblique foliation



Conclusions

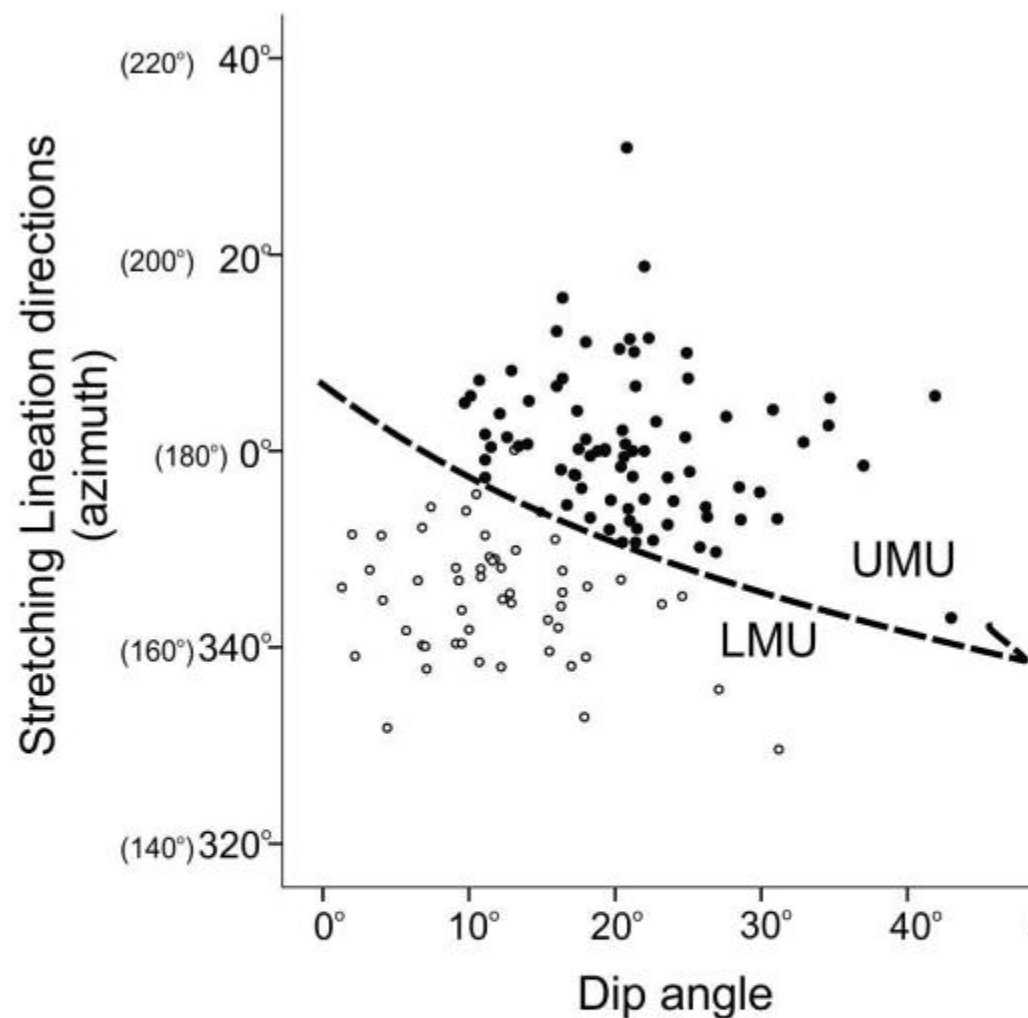
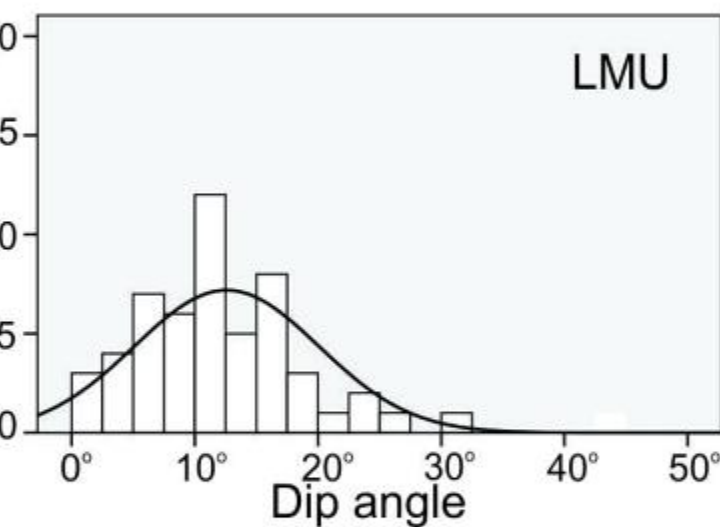
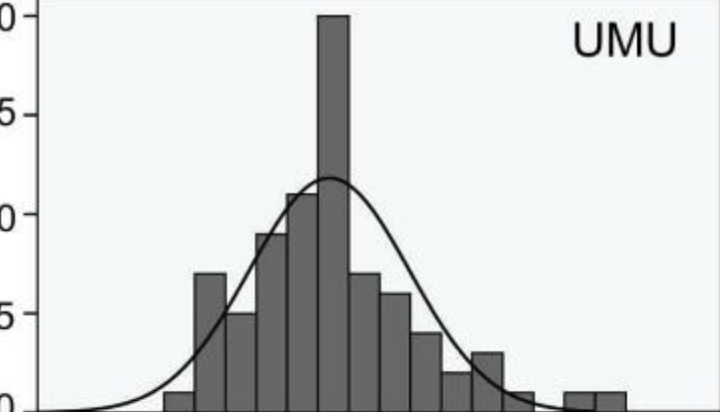
- ***Prediction map reveals two possible distinctive metamorphic units in Santorini Island***
- ***This methodology allows:***
 1. ***Quick mapping procedure***
 2. ***Identification of contacts and locations for structural and/or thermochronological sampling***

References

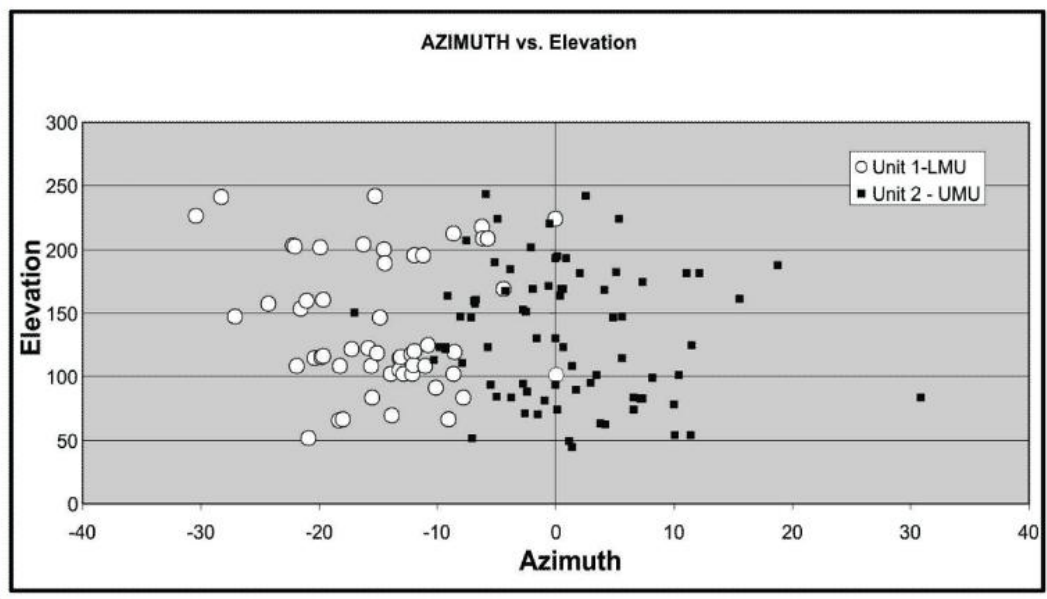
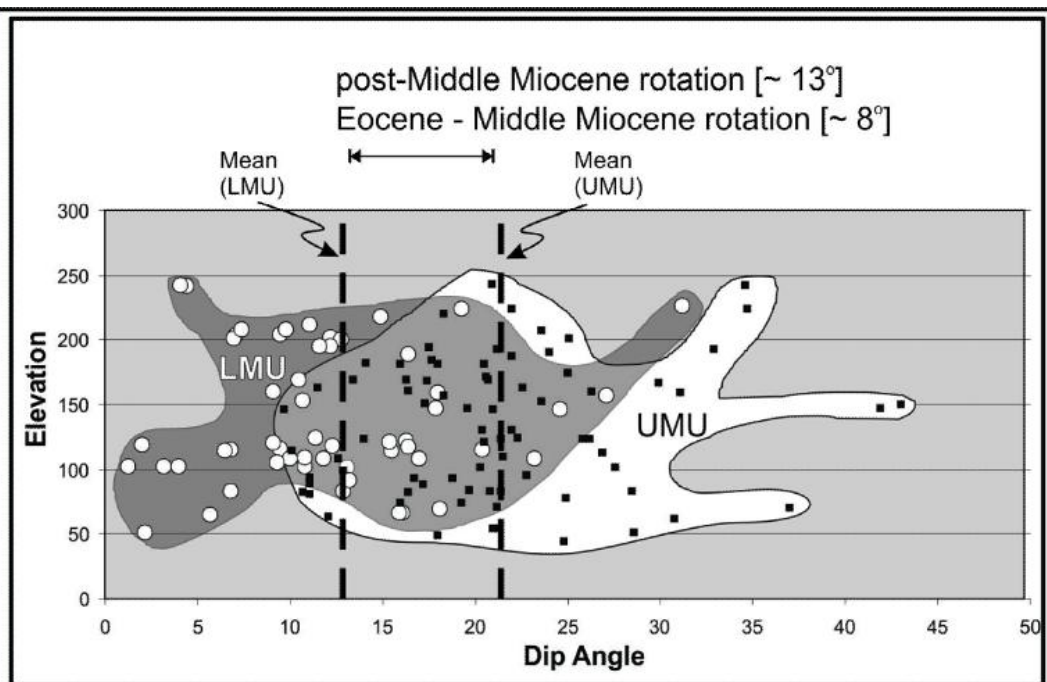
- Marsellos, A.E.**, Foster, D. A., Kidd, W.S.F., Garver, J. I., Kyriakopoulos, K., 2013. An application of GIS analysis on structural data from metamorphic rocks in Santorini Island. Hellenic Geological Society, September 2013, Crete, Greece, volume 2013, p. 1-10.
- Marsellos, A.E.**, Foster, D. A., Min, K. , Kamenov, G. D., Kidd, W.S.F., Garver, J., Kyriakopoulos, K., 2012. A structural and thermochronological study of Santorini detachment in Santorini Island, Aegean Sea. American Geophysical Union, December 2012, San Francisco, Abstr. Program.
- Marsellos, A.E.**, Kidd, W.S.F., Garver, J.I., 2010. Extension and exhumation of the HP/LT rocks in the Hellenic forearc ridge. American Journal of Science, vol. 310, p. 1-36.
- Marsellos, A.E.**, 2008. Extension and Exhumation of the Hellenic forearc, and Radiation Damage in Zircon, Ph.D. Dissertation, University at Albany.
- Marsellos, A.E.**, Kidd, W.S.F., 2008. Extension and Exhumation of the Hellenic Forearc Ridge in Kythera. J.Geology, vol. 116, p.640-651.

Thank you

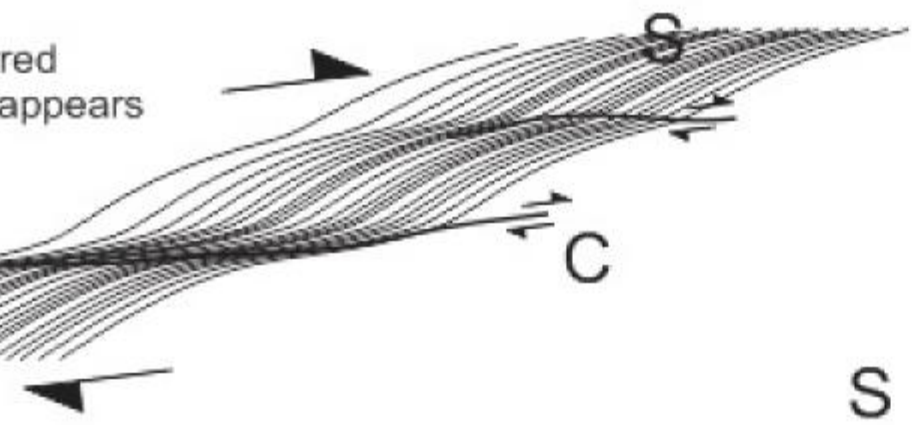




	Mean	N	Standard deviation	Range	Minimum	Maximum	Variance
UMU	21.2	78	6.9	33.3	9.7	43.0	47.8
LMU	12.0	53	6.2	29.9	1.3	31.2	39.0
Total	17.4	131	8.0	41.7	1.3	43.0	64.7

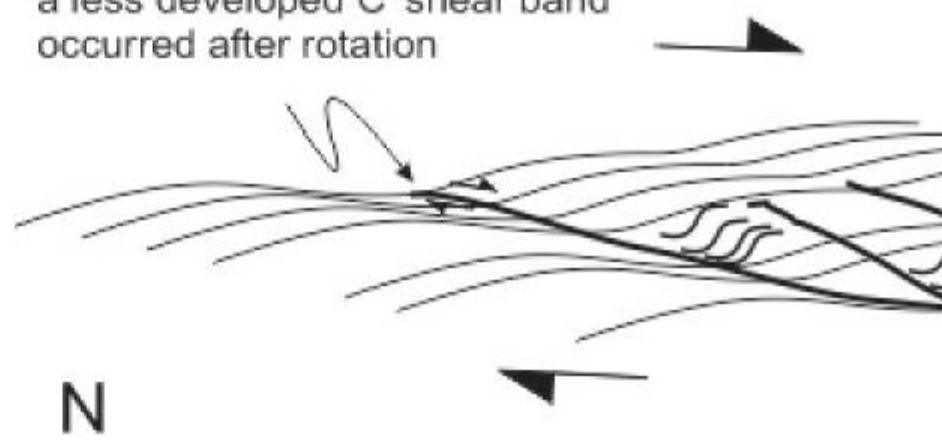


shear bands

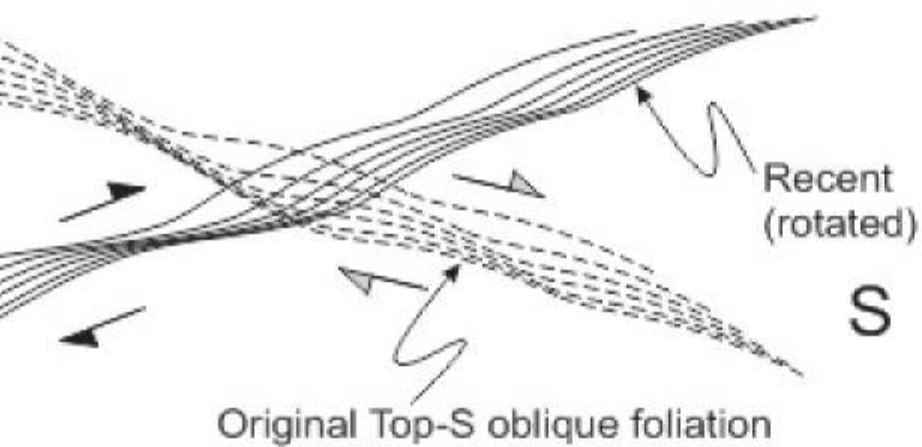


C' shear bands

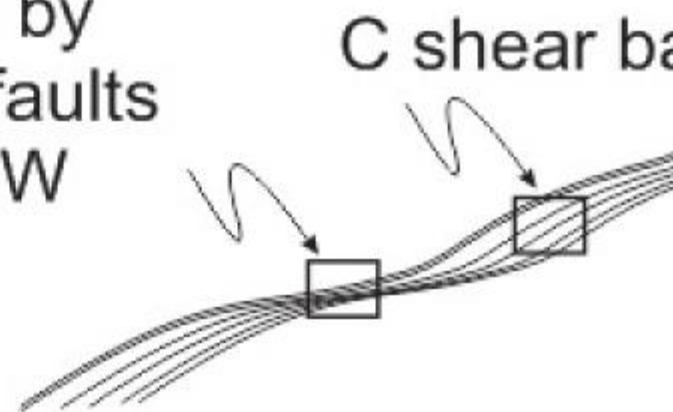
a less developed C' shear band occurred after rotation



Top-S rotated oblique foliation



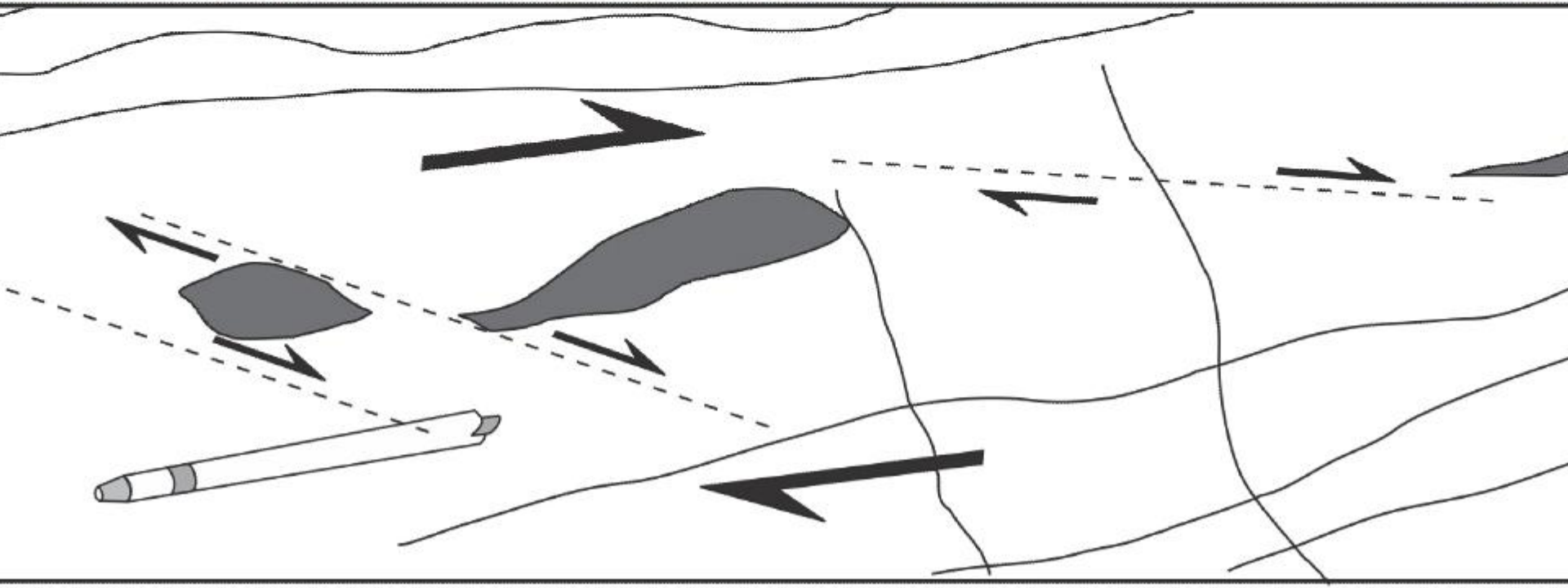
C' shear bands followed by breccia faults of Top-SW

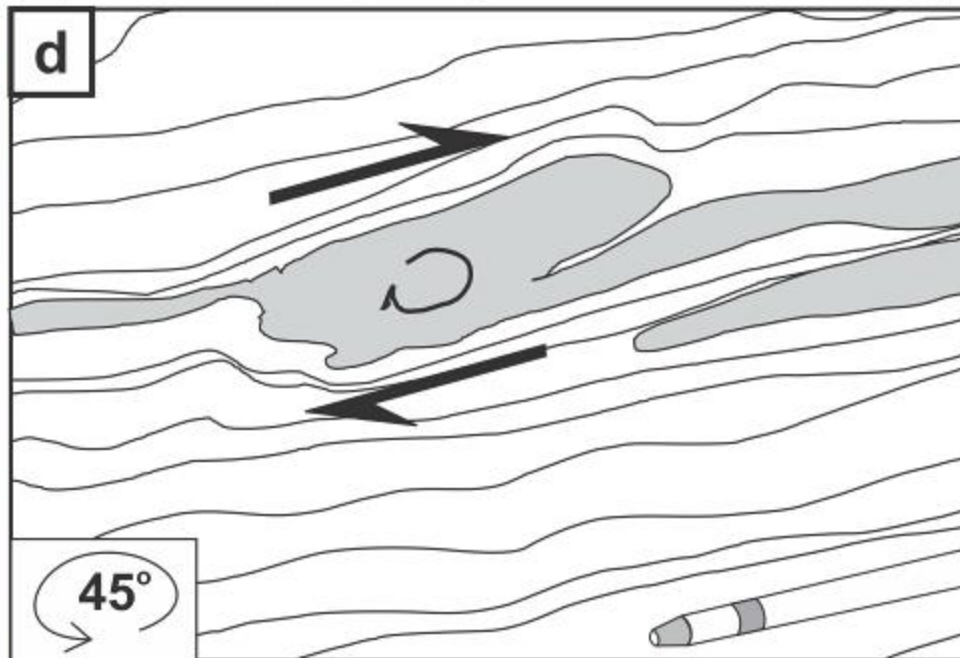
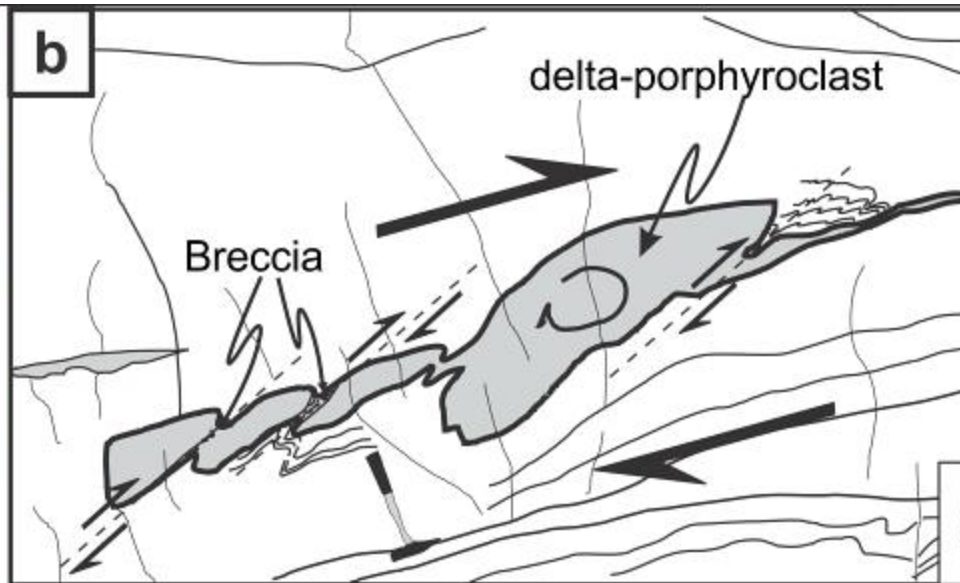


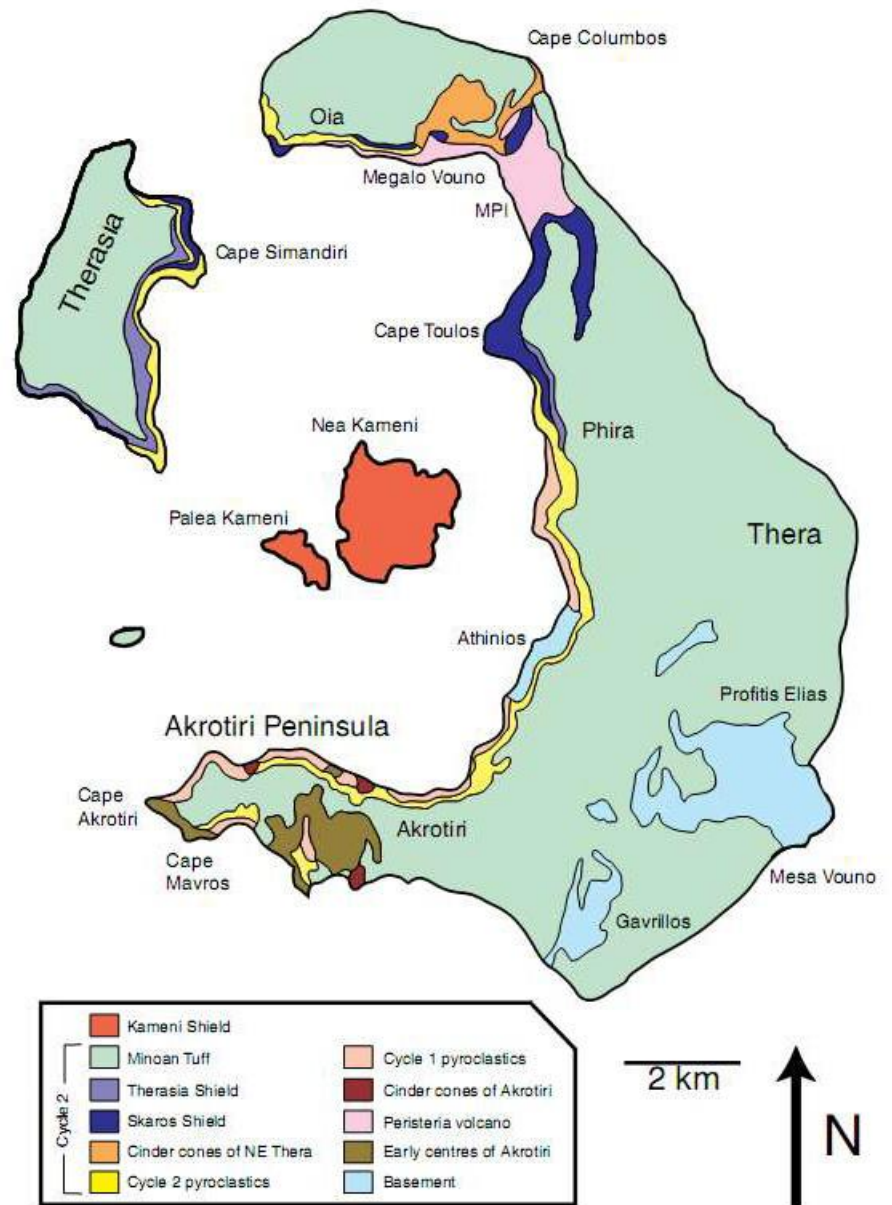
OR KYTHERA STRUCTURES SHOWING ARC
PARALLEL EXTENSION.

TO SUPPORT THE ARC NORMAL/PARALLEL
EXTENSION ON SANTORINI









Zellmer, G.F., 1995, PhD Dissertation.

Looking for a suitable study area



